Principle of antimicrobial therapy

PYPH305 Pharmacology 2 (ภกปว ๓๐๕ เภสัชวิทยา ๒)

Drugs affecting cell wall synthesis (β-lactam)

ผู้ช่วยศาสตราจารย์.ดร.ภก. อาณัฐชัย ม้ายอุเทศ

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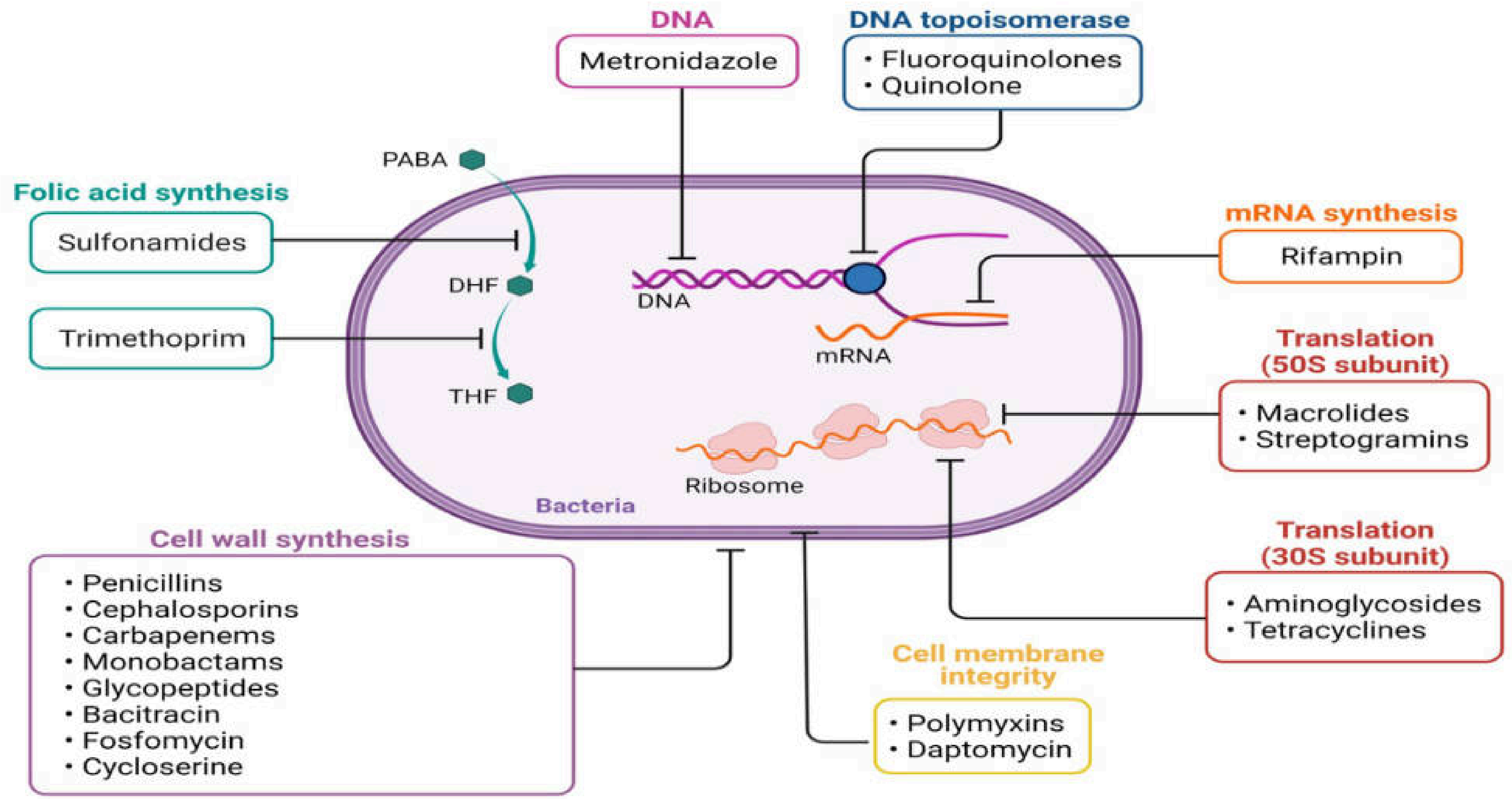
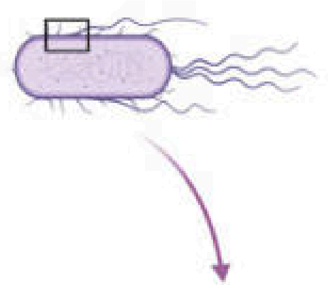
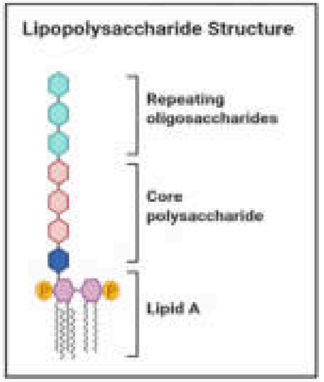
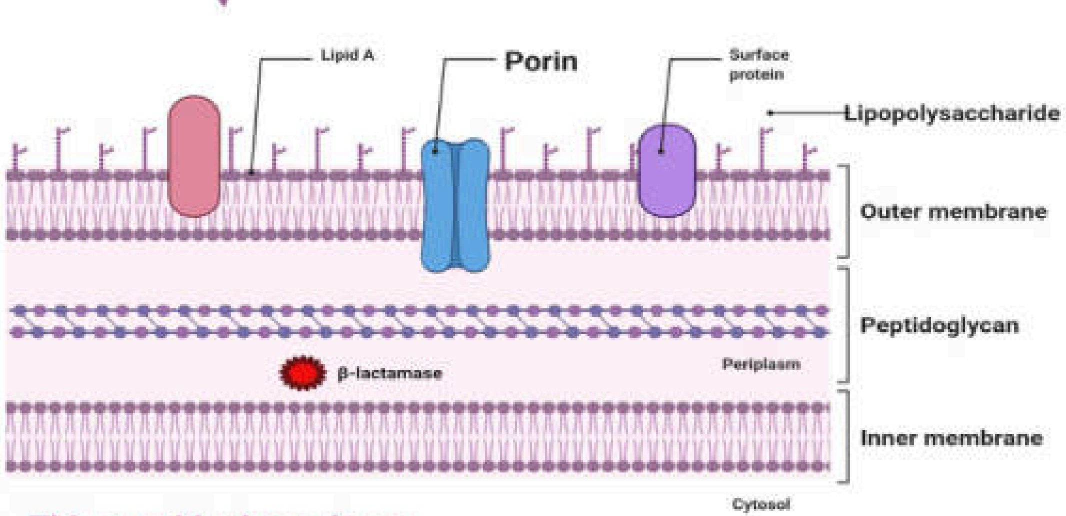

DNA
Metronidazole
DNA topoisomerase
• Fluoroquinolones
• Quinolone
PABA
Folic acid synthesis
Sulfonamides
DHF
Trimethoprim
THF
DNA
mRNA synthesis
Rifampin
mRNA
Translation
(50S subunit)
• Macrolides
• Streptogramins
Ribosome
Bacteria
Cell wall synthesis
• Penicillins
• Cephalosporins
• Carbapenems
• Monobactams
• Glycopeptides
• Bacitracin
• Fosfomycin
• Cycloserine
Translation
(30S subunit)
• Aminoglycosides
• Tetracyclines
Cell membrane
integrity
• Polymyxins
• Daptomycin

Gram negative bacteria cell wall structure

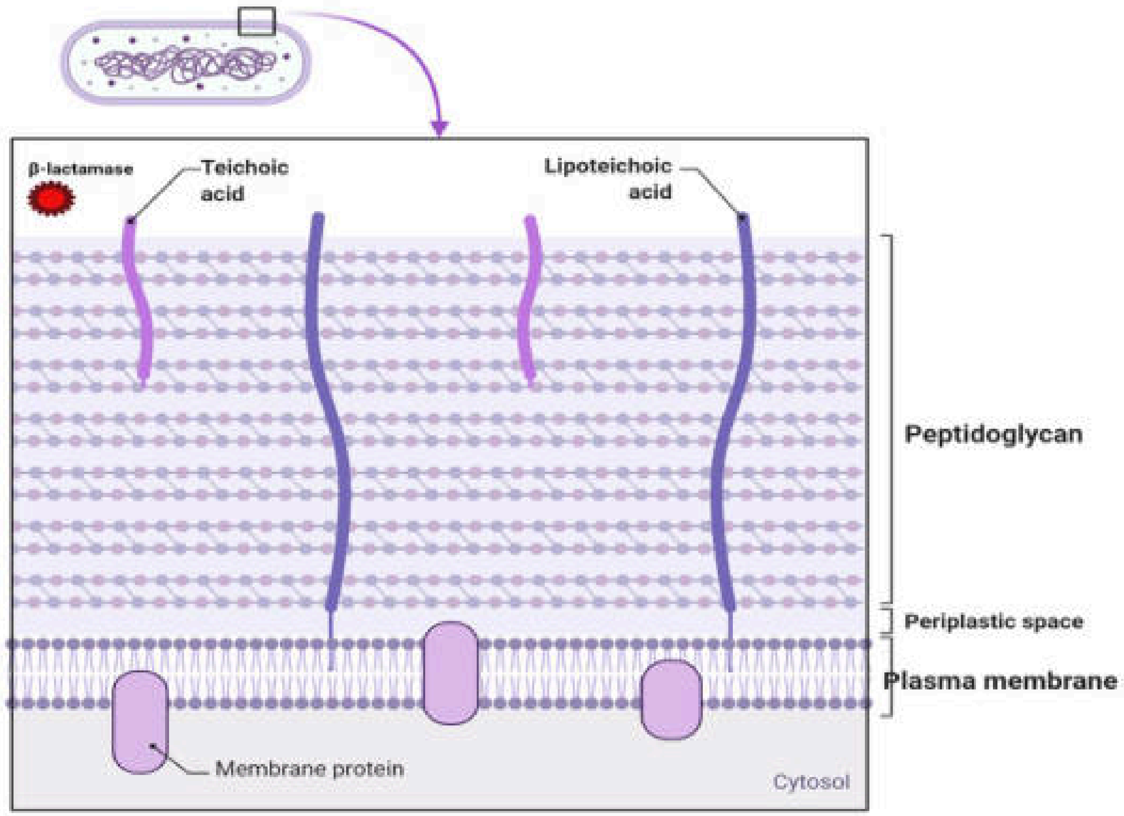




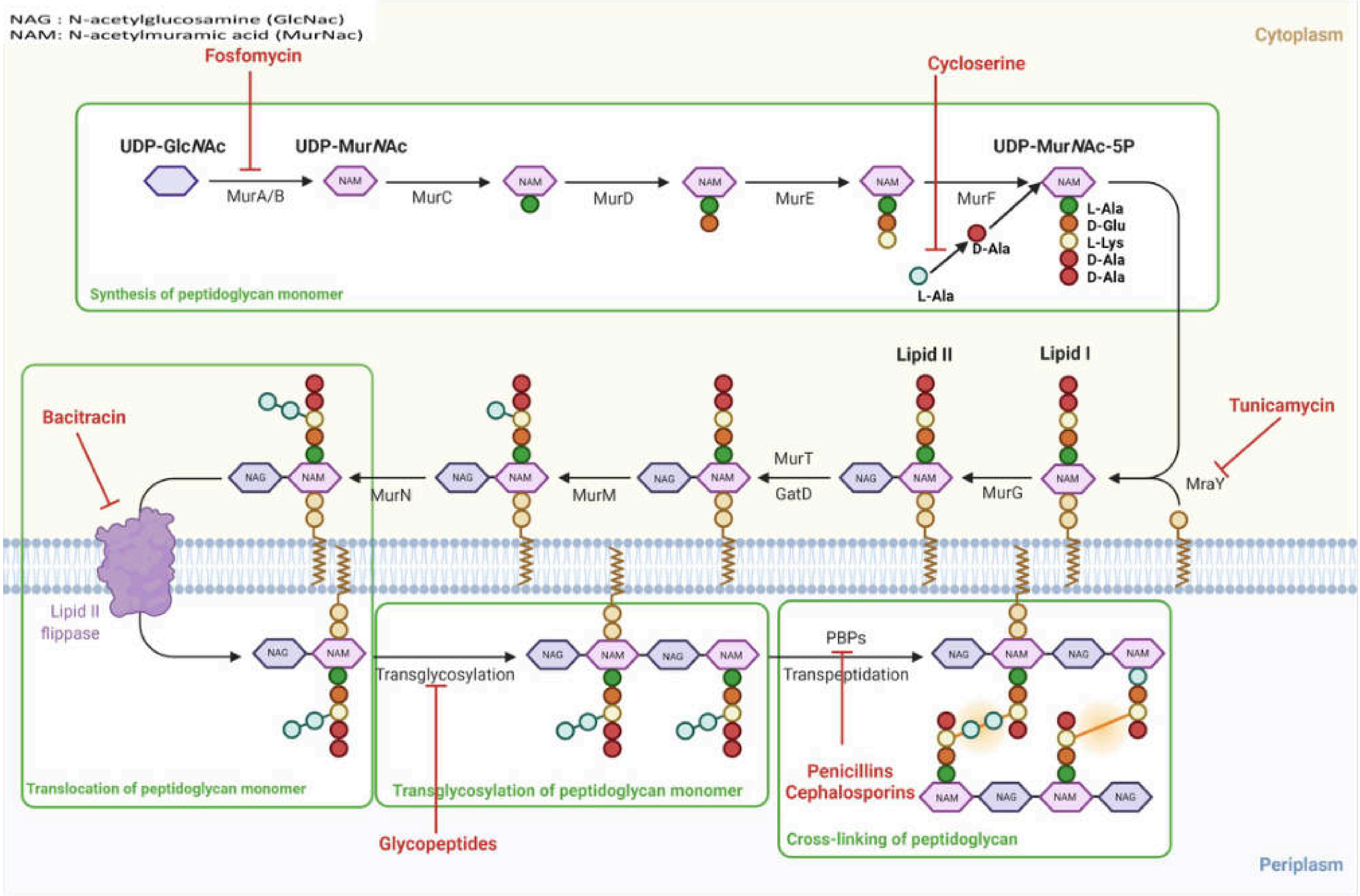


- Thin peptidoglycan layer
- Lipopolysaccharide (LPS) can form a protective barrier from antibodies and many antibiotics
- Outer membrane is also more permeable than the inner
- Porins in the outer membrane, which act like pores for particular molecules
- β-lactamase, enzyme digest β-lactam structure

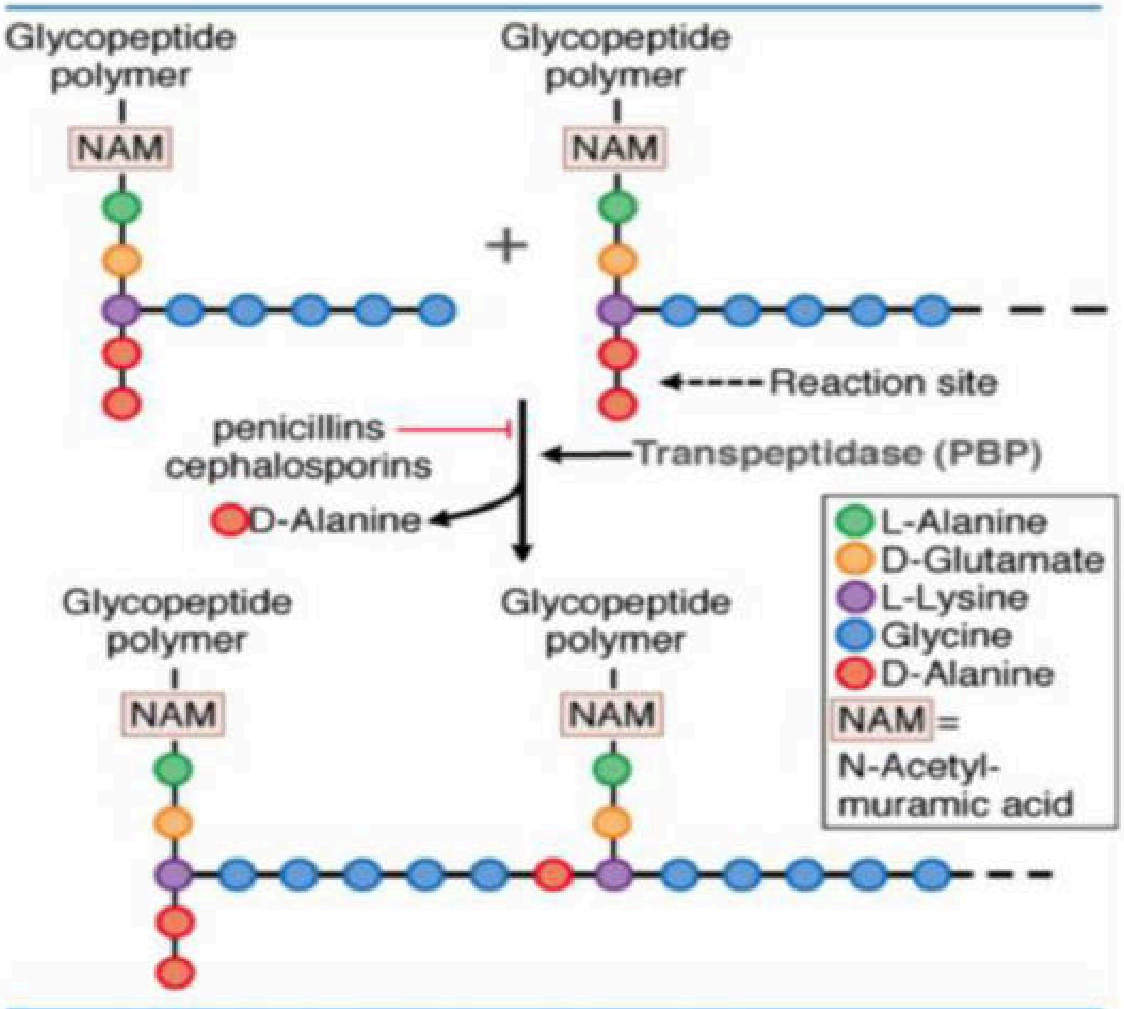
Gram-Positive Bacteria Cell Wall Structure



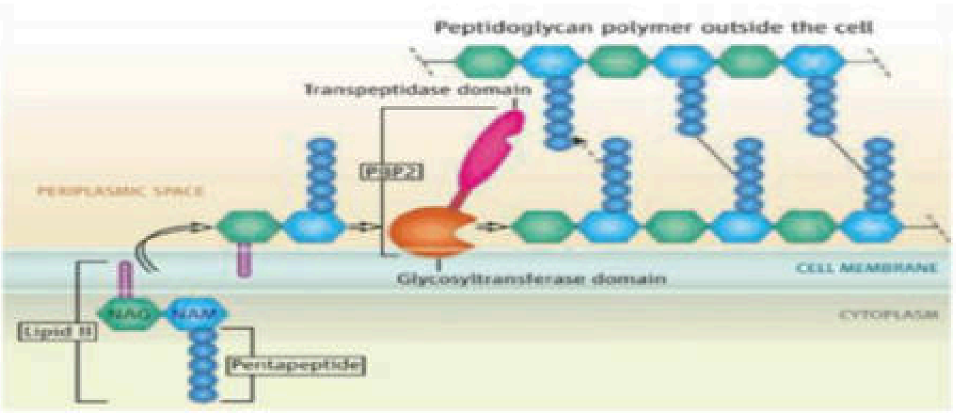
- Thick peptidoglycan layer
- Periplasmic space and it contains proteins that help the bacteria acquire nutrition



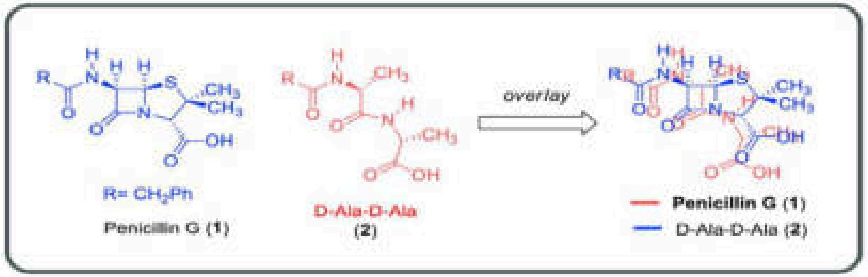
NAG : N-acetylglucosamine (GlcNac)
NAM: N-acetylmuramic acid (MurNac)
Cytoplasm
Fosfomycin
Cycloserine
UDP-GlcNAc
UDP-MurNAc
UDP-MurNAc-5P
MurA/B
MurC
MurD
MurE
MurF
D-Ala
L-Ala
L-Ala
D-Glu
L-Lys
D-Ala
D-Ala
Synthesis of peptidoglycan monomer
Lipid II
Lipid I
Tunicamycin
Bacitracin
MurN
MurM
MurT
GatD
MurG
MraY
Lipid II flippase
Transglycosylation
PBPs
Transpeptidation
Penicillins
Cephalosporins
Translocation of peptidoglycan monomer
Transglycosylation of peptidoglycan monomer
Cross-linking of peptidoglycan
Glycopeptides
Periplasm

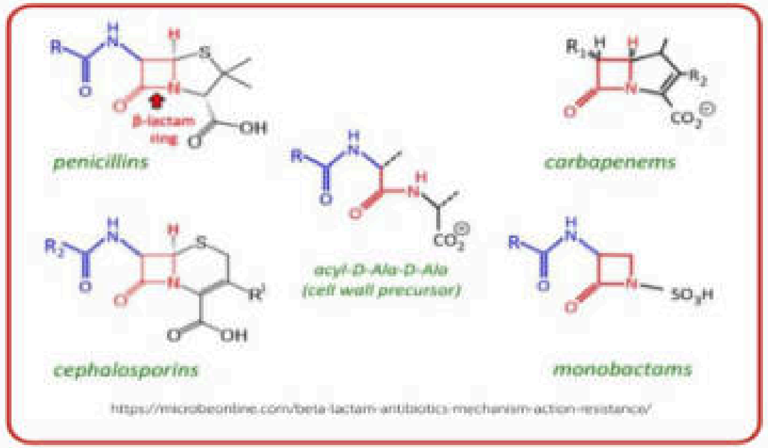


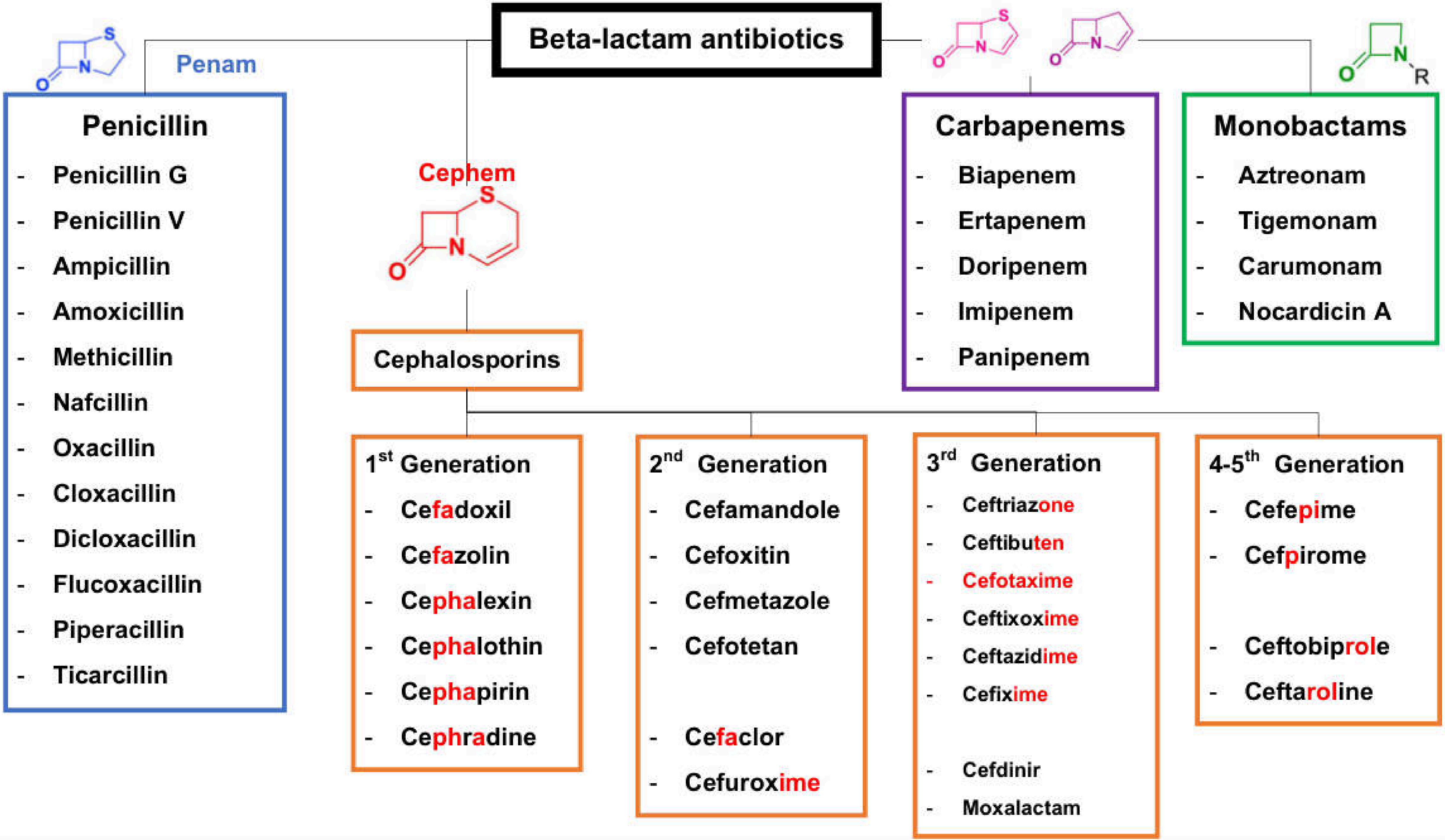
Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition*: www.accessmedicine.com



NAG : N-acetylglucosamine (GlcNac)
NAM: N-acetylmuramic acid (MurNac)







Beta-lactam antibiotics
Penam
S
N
O
Penicillin
- Penicillin G
- Penicillin V
- Ampicillin
- Amoxicillin
- Methicillin
- Nafcillin
- Oxacillin
- Cloxacillin
- Dicloxacillin
- Flucoxacillin
- Piperacillin
- Ticarcillin
Cephem
S
N
O
Cephalosporins
1st Generation
- Cefadoxil
- Cefazolin
- Cephalexin
- Cephalothin
- Cephapirin
- Cephradine
2nd Generation
- Cefamandole
- Cefoxitin
- Cefmetazole
- Cefotetan
- Cefaclor
- Cefuroxime
3rd Generation
- Ceftriazone
- Ceftibuten
- Cefotaxime
- Ceftixoxime
- Ceftazidime
- Cefixime
- Cefdinir
- Moxalactam
4-5th Generation
- Cefepime
- Cefpirome
- Ceftobiprole
- Ceftaroline
S
N
O
N
O
Carbapenems
- Biapenem
- Ertapenem
- Doripenem
- Imipenem
- Panipenem
N
O
R
Monobactams
- Aztreonam
- Tigemonam
- Carumonam
- Nocardicin A

Penicillins: Spectrum of Action

	Natural PCN (1st generation) Benzathine PCN, PCN G	Aminopenicillins (2nd generation) Amoxicillin, Ampicillin	Anti-*Staph* PCNs (3rd generation) Nafcillin, Dicloxacillin	Anti-*Pseudomonal* PCNs (4th generation) Piperacillin + tazobactam
MSSA	+	+ (Only if combined with a β-lactamase inhibitor)	++	+
MRSA	-	-	-	-
Strep (excl. Viridans)	++	+	+	+
Enterococcus	+	++	-	+
Gram negatives	-	+ (Only if combined with a β-lactamase inhibitor)	-	++
Anaerobes	+	++	-	++
Pseudomonas	-	-	-	++

??

R side chain of pen G and consists of a hydrophobic benzene ring.

Gr-neg bacteria, have porins in their outer membranes that do not allow to the periplasmic space.

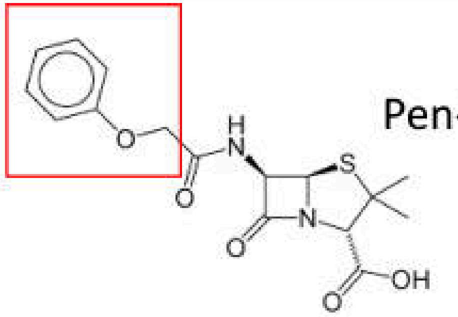
An additional amino group in their side chain increases their hydrophilicity and allows them to pass through the porins in the outer membranes of some enteric gram-negative rods

Bulky residues on their R side chains prevent binding by the staphylococcal B-lactamases

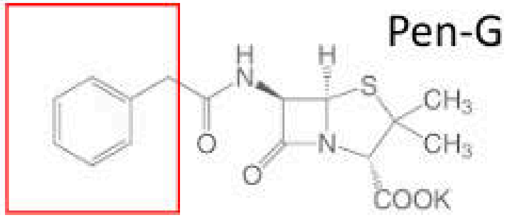
The side chain of piperacillin is polar, which increases its ability to pass through the outer membrane porins of some gram-negative bacteria

ทำไม Penicillin แต่ละชนิด จึงมีความไวต่อเชื้อแตกต่างกัน

กลุ่ม	ชื่อยา	ขอบเขตการออกฤทธิ์
Natural penicillin	Penicillin G Penicillin V	Narrow spectrum - Streptococcus spp., MSSA - Anaerobes
Penicillinase-resistance penicillin	Methicillin Nafcillin Oxacillin Cloxacillin Dicloxacillin	Narrow spectrum - Staphylococcus aureus - Staphylococcus epidermidis - +penicillinase



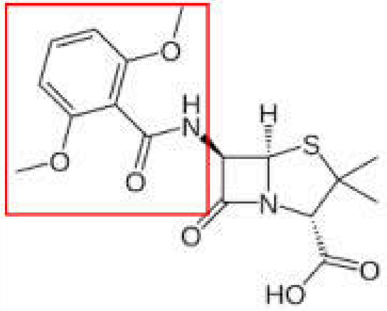
Pen-V



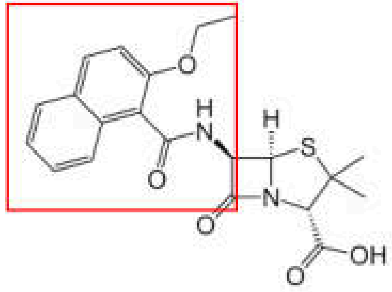
Pen-G

R side chain of Pen G ,V and consists of a hydrophobic benzene ring, Do not allow pass through porins.

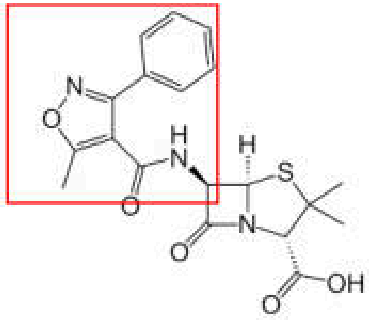
Methicillin



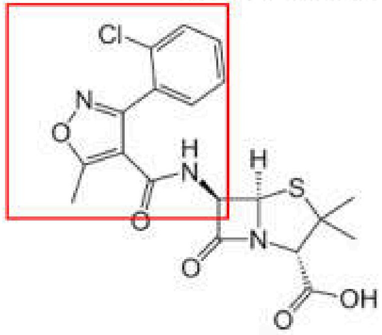
Nafcillin



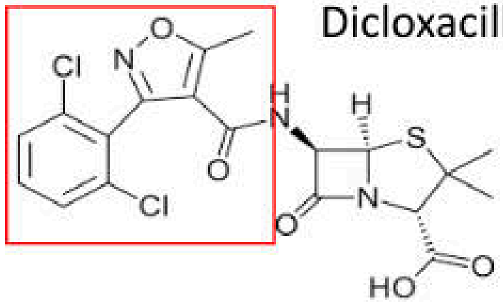
Oxacillin



Cloxacillin



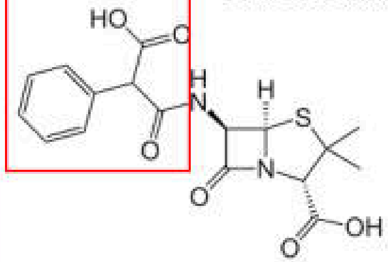
Dicloxacillin



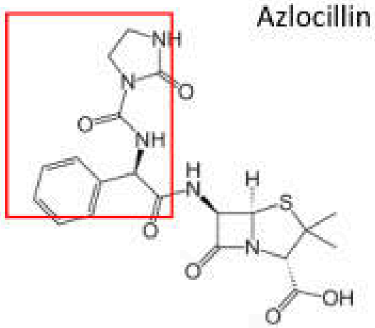
Bulky residues on their R side chains that prevent binding by the staphylococcal B-lactamases

กลุ่ม	ชื่อยา	ขอบเขตการออกฤทธิ์
Aminopenicillins	Ampicillin Amoxicillin Bacampicillin Ciclacillin	Broad spectrum - Listeria monocytogenes - Heaemophilus influenzae - Escherichia coli - Neisseria gonorrhoeae
Antipseudomonal penicillins	Carbenicillin Ticarcillin Temocillin Azlocillin Mezlocillin Piperacillin	Broad spectrum เหมือนกลุ่ม aminopenicillin และ Pseudomonas aeruginosa Enterobacter spp. Klebsiella spp. Anaerobes

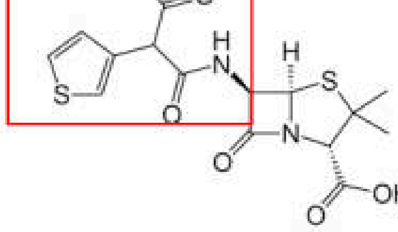
Carbenicillin



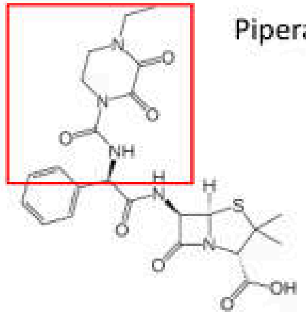
Azlocillin



Ticarcillin

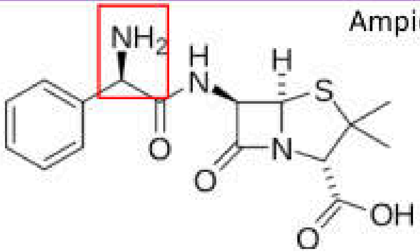


Piperacillin

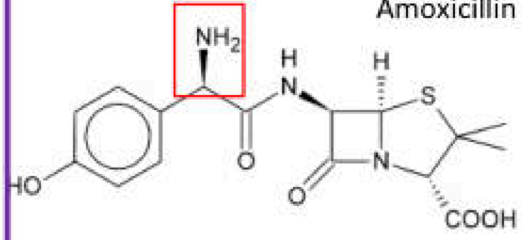


The side chain is polar, increases its ability to pass through the porins of some gram-negative bacteria

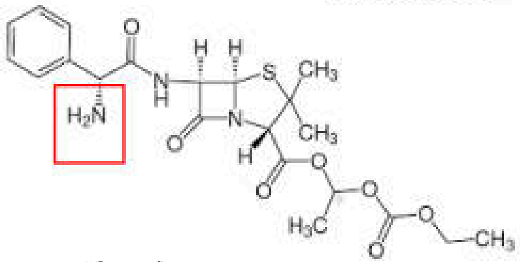
Ampicillin



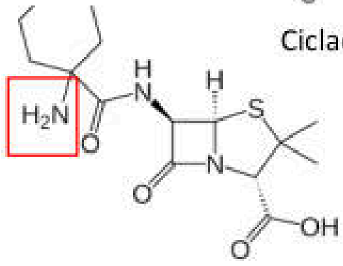
Amoxicillin



Bacampicillin



Ciclacillin



An additional amino group increases hydrophilicity allows them to pass through the porins of some gram-negative

Penicillin Binding Proteins (PBPs)

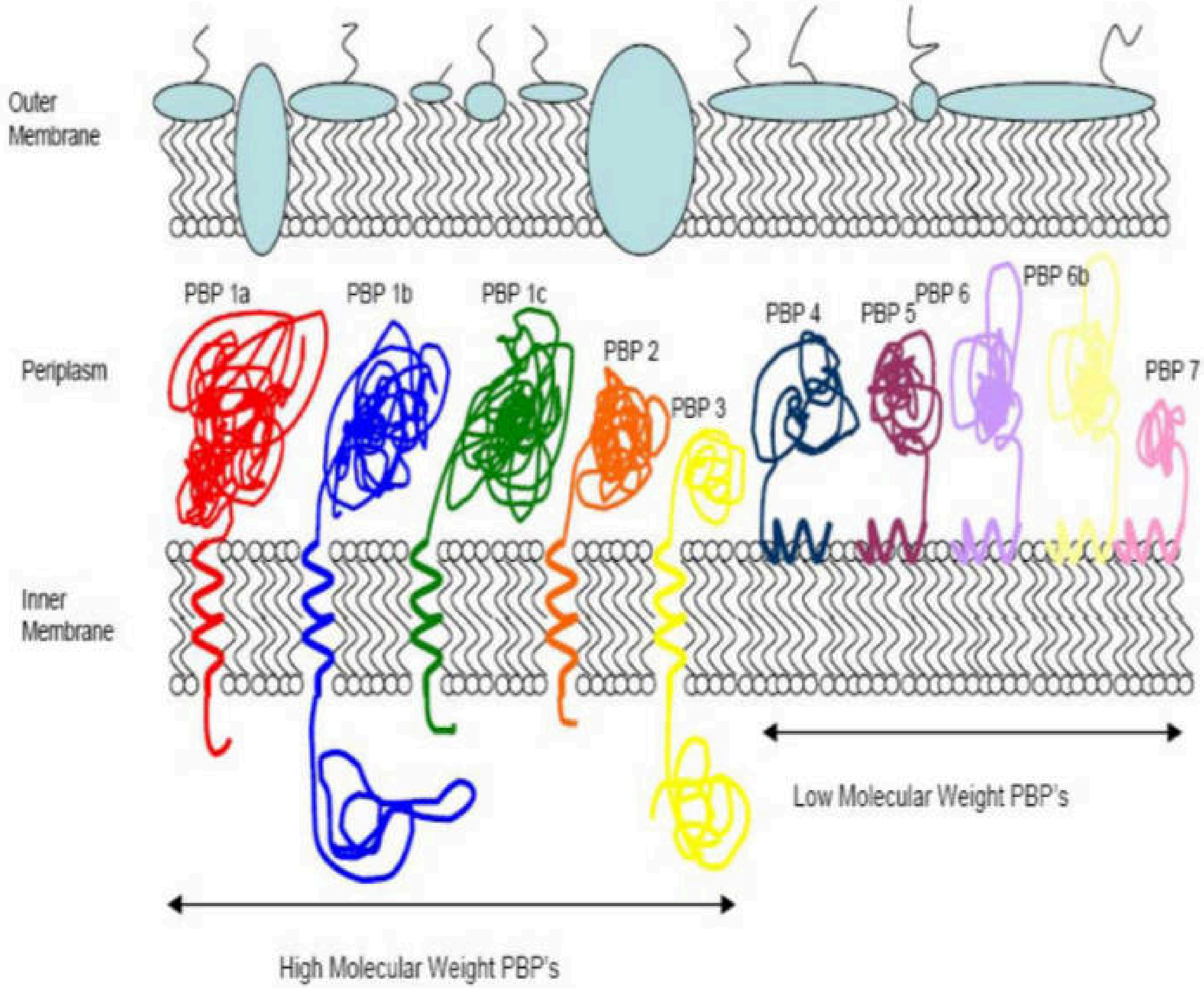
High molecular weight mass PBP

Transpeptidase activity (c-terminal of PBP domain)

- Class A
  - N-terminal: glycosyltransferase
  - N-terminal: transpeptidase
- Class B
  - N-terminal: transpeptidase

Low molecular weight mass PBP

- Class C
  - Cell separation
  - Peptidoglycan maturation and recycle



Structure and localization of the penicillin binding proteins

The high molecular weight PBP's, PBP1a, PBP1b, PBP1c, PBP2, and PBP3 are all attached to the inner membrane through a transmembrane helix with their enzymatic domain pointing into the periplasm. PBP1a, 1b, and 1c have two enzymatic domains, a transpeptidase and a transglycosylase domain.
PBP 2 and PBP 3 have a single transpeptidase domain.

The low molecular weight PBP's are attached to the inner membrane through an amphipathic helix.
One enzymatic domain with endopeptidase (PBP4,and 7),
or carboxypeptidase (PBP5, 6, and 6b).

	Class A							Class B										Class C			
	A1	A2	A3	A4	A5	A6	A7	B1	B2	B3	B4	B5	B5	B6	B-like-I	B-like-II	B-like-III	Type-4	Type-5	Type-7	Type-AmpH
Gram -																					
Escherichia coli K12	PBP1a *ponA*	PBP1b *ponB*				PBP1c *pbpC*	MGT *mgt*		PBP2 *pbpA*	PBP3 *ftsI*								PBP4 *dacB*	PBP5 *dacA*; PBP6 *dacC*; PBP6b *dacD*	PBP7 *pbpG*	PBP4b *yefw*; AmpH *ampH*
Neisseria gonorrhoeae FA 1090	PBP1 *ponA*									PBP2 *ftsI*								PBP3 *pbp3*		PBP4 *pbp4*	
Gram +								PBP5fm													
Bacillus subtilis 168			PBP1 *ponA*	PBP2c *pbpF*	PBP4 *pbpD*		PBP2d *pbpG*	PBP3 *pbpC*		SpoVD *spoVD*	PBP2b *pbpB*	PBP2a *pbpA*	PbpH *pbpH*	PBP4b *yrrR*				PBP4a *dacC*	DacF *dacF*; PBP5 *dacA*; PBP5* *dacB*		PBP4* *pbpE*; PbpX *pbpX*
Staphylococcus aureus MRSA252			PBP2 *pbp2*				MGT *mgt*	PBP2a *mecA*			PBP1 *pbpA*	PBP3 *pbp3*							PBP4 *pbp4*		
Listeria monocytogenes 4b F236			PBP1 *lmo1892*	PBP4 *lmo2229*				PBP *lmo0441*			PBP2 *lmo2039*	PBP3 *lmo1438*							PBP5 *lmo2754*		PBP *lmo0540*
Enterococcus faecalis V583			PBP1a *EF_1148*	PBP2a *EF_0680*	PBP1b *EF_1740*			PBP4 *EF_2476*			PBP2 *EF_0991*	PBP2b *EF_2857*							DacF *EF_3129*		PBP *EF_0746*
Streptococcus pneumoniae R6			PBP1a *pbpA*	PBP2a *pbp2a*	PBP1b *pbp1b*						PBP2x *pvpX*	PBP2b *pbp2b*							PBP3 *pbp3*		
Actinomycetes																		R39		K15	R61
Streptomyces coelicolor A3(2)							*3 PBP-A* *sco3901* *sco2897* *sco5039*		PBP2 *sco2608*	PBP3 *sco2090*					4 PBP-B *sco3771* *sco3156* *sco4013* *sco3847*		*3 PBP-B* *sco3157* *sco3771* *sco3156*	PBP4 *sco3408*; PBP4 *sco6131*	PBP *sco4439*	PBP7 *sco3811*; PBP *sco4847* *sco7050*	PBP *sco0830*; PBP *sco7561*; PBP *sco2283*
Mycobacterium tuberculosis H37Rv				PBP1 *ponA1*			PBP1A (r) *ponA2*	PBPA *pbpA*		PBP2 *pbpB*						PBP-lipo *Rv2864c*		PBP4 *Rv3627*	PBP5 *dacB1*	PBP7 *dacB2*	PBP *Rv0907*; PBP; PBP *Rv1367c*
Cyanobacteria																					
Anabaena species PCC7120	PBP1 *all2952*	3-4-5-6 *alr4579* *alr5324* *alr5326* *all2981*				PBP2 *alr5101*			PBP7 *alr5045*	PBP8 *alr0718*								PBP10 *alr1666*; PBP11 *alr0054*			PBP9 *alr0153*; PBP12 *all2656*

PBPs classification. Complete set of PBPs from 10 bacteria

Eric Sauvage., et al. FEMS Microbiol Rev 2008;32:234–258

High molecular weight PBPs of organisms that resist b-lactams by expressing low affinity PBPs

	Class A (bifunctional)			Class B (monofunctional)		
Staphylococcus aureus	PBP2			PBP1	PBP3	**PBP2a** ↗ *mecA*
Enterococci	PBP1a *ponA*	PBP1b *pbpZ*	PBP2a *pbpF*	PBPC(B) *pbpB*	PBP2b *pbpA*	PBP5 ↗
Streptococcus pneumoniae	*PBP1a*	PBP1b	PBP2a	*PBP2x*	*PBP2b*	
Neisseria	PBP1 *ponA*			*PBP2* *penA*		
Haemophilus influenzae	PBP1a	PBP1b		PBP3 *ftsI*	PBP2	
Helicobacter pylori	PBP1a			PBP3	PBP2	

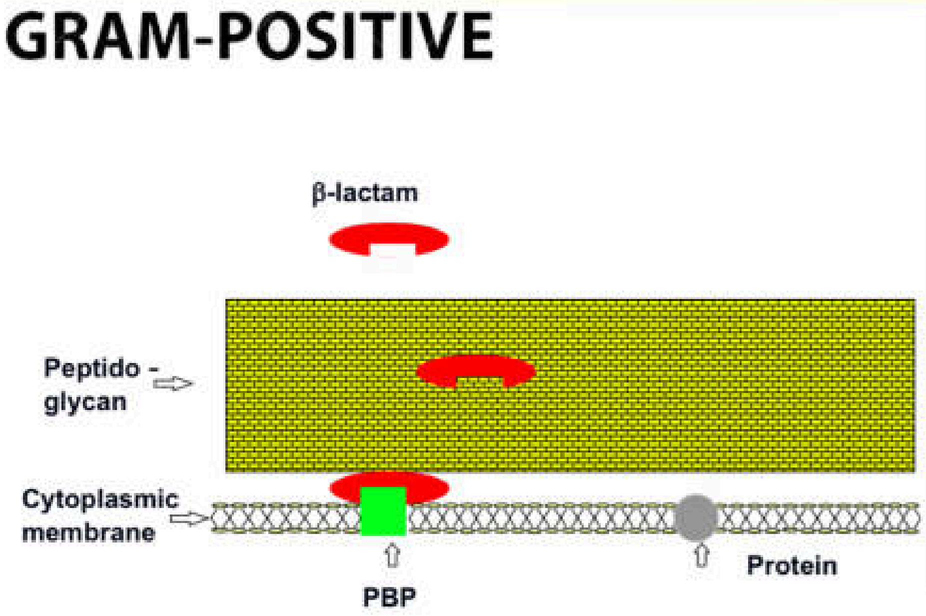
Low-affinity PBPs are underlined. Discontinuous underline indicate intrinsic low affinity. Italic characters indicate mosaicity. Nonitalic characters indicate point mutations. An arrow indicate overexpression. Bold characters indicate acquisition of exogenous origin. Alternative gene names are given below their respective product.

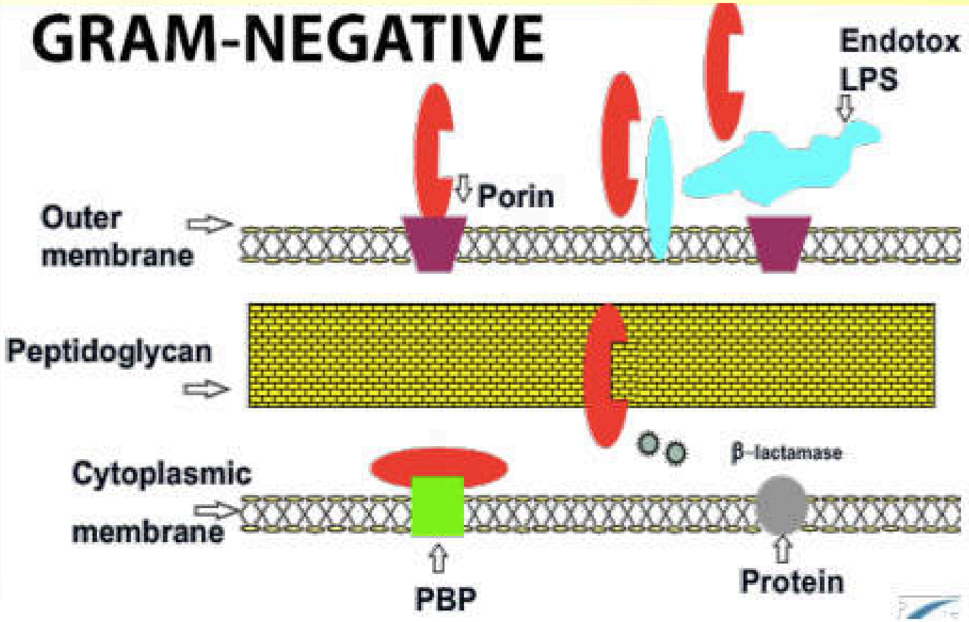
Andr´e Zapun., et al. FEMS Microbiol Rev 2008;32:361–385

IC50s of -lactams for *S. pneumoniae* IU1945

β-Lactam class and drug	MIC (μg/ml)	IC_{50} (μg/ml)[a]						PBP selectivity
		PBP1a	PBP1b	PBP2x	PBP2a	PBP2b	PBP3	
Penicillins								
Penicillin V	0.008	0.022 ± 0.001	0.036 ± 0.005	0.0074* ± 0.0009	0.14 ± 0.03	0.15 ± 0.05	0.0046 ± 0.0003	2x, 3
Penicillin G	0.008	0.020 ± 0.001	0.028 ± 0.002	0.0059* ± 0.0000	0.038 ± 0.002	0.13 ± 0.01	0.0024 ± 0.0001	2x, 3
Ampicillin	0.016	0.077* ± 0.031	0.24 ± 0.11	0.046* ± 0.031[b]	0.14 ± 0.09[b]	0.12 ± 0.05	0.022 ± 0.013[b]	1a, 2x, 3[d]
Amoxicillin	0.016	0.069 ± 0.029	0.17 ± 0.12[b]	0.025* ± 0.014	0.12 ± 0.05	0.078 ± 0.027	0.012 ± 0.005	2x, 3
Oxacillin	0.016	1.5 ± 0.3	1.3 ± 0.2	0.038 ± 0.013	3.7 ± 0.6	1.4 ± 0.3	0.071* ± 0.006	2x, 3
Piperacillin	0.016	4.8 ± 4.0[b]	4.1 ± 3.5[b]	0.020 ± 0.008	0.94 ± 0.07	0.18 ± 0.11[b]	0.056* ± 0.000	2x, 3[d]
Methicillin	0.125	9.9 ± 0.4	4.9 ± 0.5	0.083 ± 0.012	4.3 ± 0.9	2.5 ± 0.4	0.16* ± 0.00	2x, 3
Amdinocillin	4	73* ± 2	48 ± 4	>1,000	>1,000	>1,000	>1,000	1a, 1b
6-APA	64	>1,000	>1,000	>1,000	>1,000	>1,000	>1,000	NS

β-lactams: Resistance





The peptidoglycan layers of Gram(+) allow the diffusion of β-lactams to entry and bind target PBP's on the outer surface of the cytoplasmic membrane

Natural Resistance

- PBP's *Enterococcus* are different from other Gram(+) (higher lipid content in cell wall) which causes a resistance Penicillins and 1st – 5th gen Cephalosporins

Natural Resistance

- Many Gram(-) are naturally resistant to penicillin G and oxacillin because the β-lactams is prevented from entering the cell by the LPS which blocks the porins.

Antibiotic-resistant bacterial pathogens

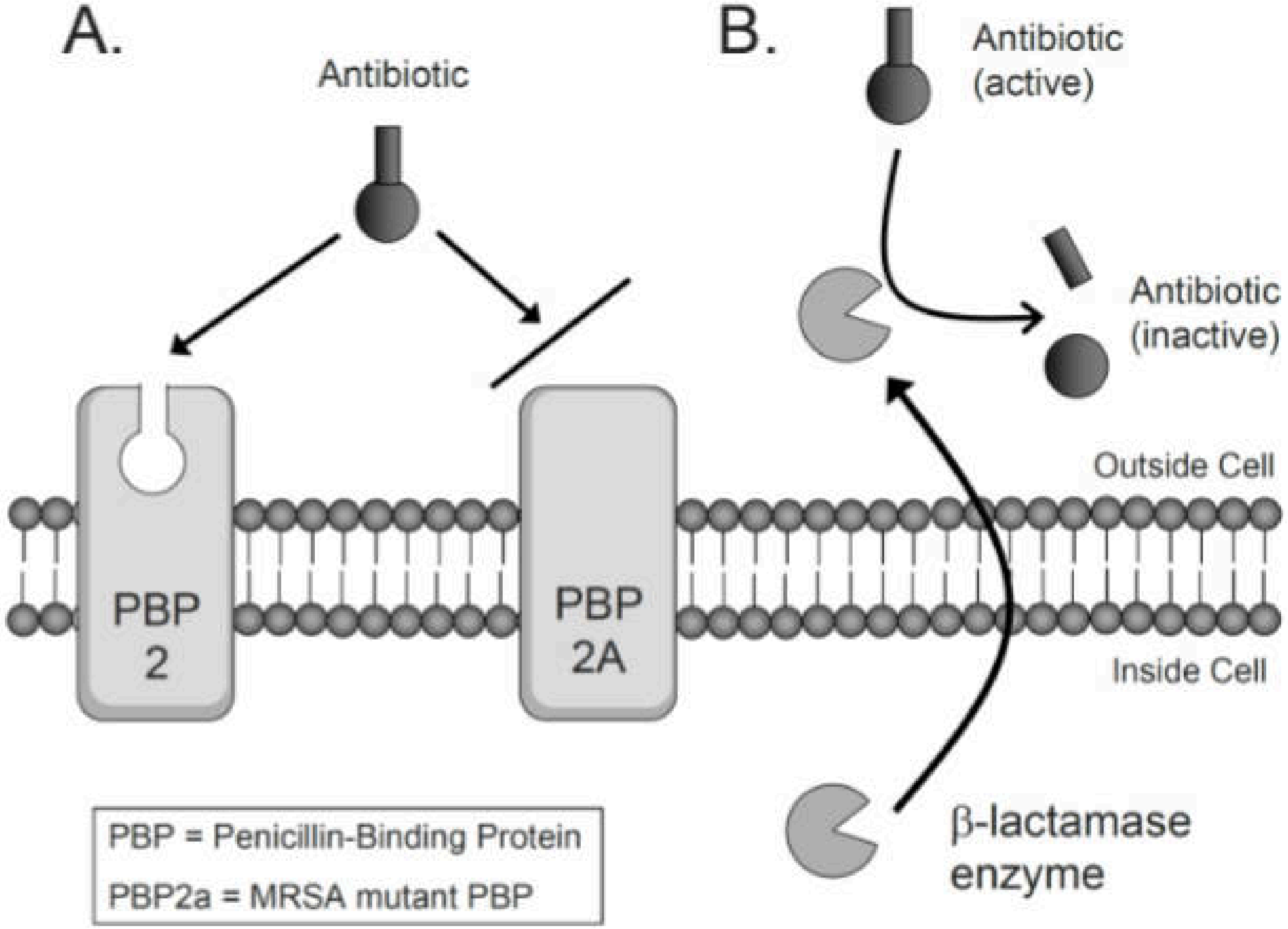


Diagram of the two principal antibiotic resistance mechanisms observed in MRSA bacteria.

A. Expression of alternate form of penicillin-binding protein PBP2, called PBP2a (*mecA*), with reduced binding affinity for antibiotic.

B. Production and release of β - lactamase enzyme which cleaves and inactivates antibiotic molecules.

β-lactams: Resistance

	Class A (bifunctional)			Class B (monofunctional)		
Staphylococcus aureus	PBP2			PBP1	PBP3	PBP2a ↗
						mecA
Enterococci	PBP1a	PBP1b	PBP2a	PBPC(B)	PBP2b	PBP5 ↗
	ponA	pbpZ	pbpF	pbpB	pbpA	
Streptococcus pneumoniae	PBP1a	PBP1b	PBP2a	PBP2x	PBP2b	
Neisseria	PBP1			PBP2		
	ponA			penA		
Haemophilus influenzae	PBP1a	PBP1b		PBP3	PBP2	
				ftsI		
Helicobacter pylori	PBP1a			PBP3	PBP2	

Low-affinity PBPs are boxed. *Hatched borders* indicate an intrinsic low affinity. An *arrow* indicate overexpression. *No shading* indicates point mutations, *light shading* indicates mosaicity, and *dark shading* indicates acquisition of exogenous origin. Alternative gene names are given below their respective product

β-lactams: Resistance

1. **Enzymatic inactivation** (most common)
   - **β-lactamase** (produced by *S.aureus, H.influenzae*, *E.coli)*:
     Penicillins >> **Cephalosporins**
   - Other β-lactamases e.g. **Extended–Spectrum β-lactamases (ESBLs)** (*P.aeruginosa*, *Enterobacter* spp): both **Penicillins** & **Cephalosporins**
2. **Natural resistance:** impaired penetration of drug to target PBPs
   - Gram(-) → impermeable outer membrane of cell wall
3. **Alteration of target site PBPs:** low affinity for β-lactams binding
   - Methicillin resistance in staphylococci (MRSA)
   - Penicillin resistance in pneumococci and enterococci
4. **Efflux**
   - Gram(-) → produce an efflux pump which transport β-lactams out across cell wall outer membrane

β-lactamase Inhibitors

Beta-lactamase inhibitors (BLI) have a beta(ß)-lactam ring, but have weak or poor antibacterial activity.

- They have a very high affinity for ß-lactamases
- They act as a trap, and are hydrolyzed in preference to the ß-lactam drug. The drug is left intact to act on the bacteria (cell wall).
- Should be called penicillinase inhibitors, because they are active against:
    - Staph penicillinase
    - Penicillinase of *K. pneumoniae*
    - ESBL (to a greater or lesser degree) - if the penicillinase is being overproduced, the inhibitor effect may be diluted (Inoculum Effect)

Combination β-lactams - β-lactamase inhibitor:

Amoxicillin + Clavulanic Acid
Ampicillin + Sulbactam
Ticarcillin + Clavulanic Acid
Piperacillin + Tazobactam

Never Alone!

NEW ISSUE - BLI can act as inducers and actually stimulate enzyme (beta-lactamase) production. It is possible to see the following:

Pseudo monas	Ticarcillin = S	Ticarcillin/Clavulanic = R
Enterobacteriaceae	Piperacillin = S	Piperacillin/Tazobactam = R

Penicillins/β-lactamase inhibitors: Clinical use

Drug	Clinical use or target pathogens	Recommended dose	Important ADRs
Amoxicillin/clavulanate (IV, PO)	- Streptococci, MSSA - Enterococci - Enterobacteriaceae - P. aeruginosa (only pip/tazo) - A. baumannii (only amp/sul, pip/tazo) - Anaerobes	IV: 1000/200 mcg q 8h PO: 500/125 mg tid or 875/125 mg bid	- Like ampicillin, amoxicillin - Hepatotoxicity
Ampicillin/sulbactam (IV)		2/1 g q 6h	- Like ampicillin, amoxicillin
Piperacillin/tazobactam (IV)		4.5g q 6-8 h (q 6h กรณีสงสัย P. aeruginosa)	- Thrombocytoprnia

Penicillins: Pharmacokinetics

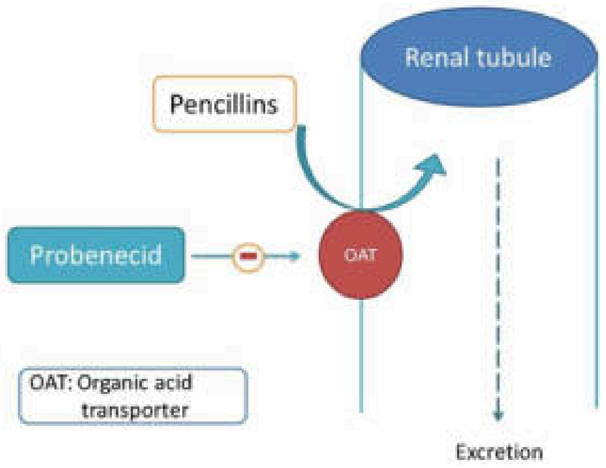
- **Absorption:**
  - **Acid-stable**
    - **Aminopenicillin: Ampicillin, Amoxicillin**
    - **Penicillinase resistance penicillin: Methicillin, Nafcillin, Oxacillin, Cloxacillin, Dicloxacillin, Flucloxacillin**
  - **Protein binding**
    - **Nafcillin, Oxacillin, Cloxacillin, Dicloxacillin, Flucloxacillin**

      : Highly protein-bound (90-97%)
    - **Penicillin G, Ampicillin**

      : Less protein-bound
  - **Food interferes drug absorption** → 1– 2 hours before or after a meal
    - **Amoxicillin** (Exception)
  - **Delay absorption** → prolonged blood conc.
    - **Benzathine** & **Procaine Penicillins** (IM)

Penicillins: Pharmacokinetics

- **Distribution:**
  - Penicillin conc. in tissues are equal to those in serum
  - Limited penetration into eye, prostate or across BBB
- **Excretion**
  - Mainly excreted by **kidneys** (renal failure → prolong $T_{1/2}$)



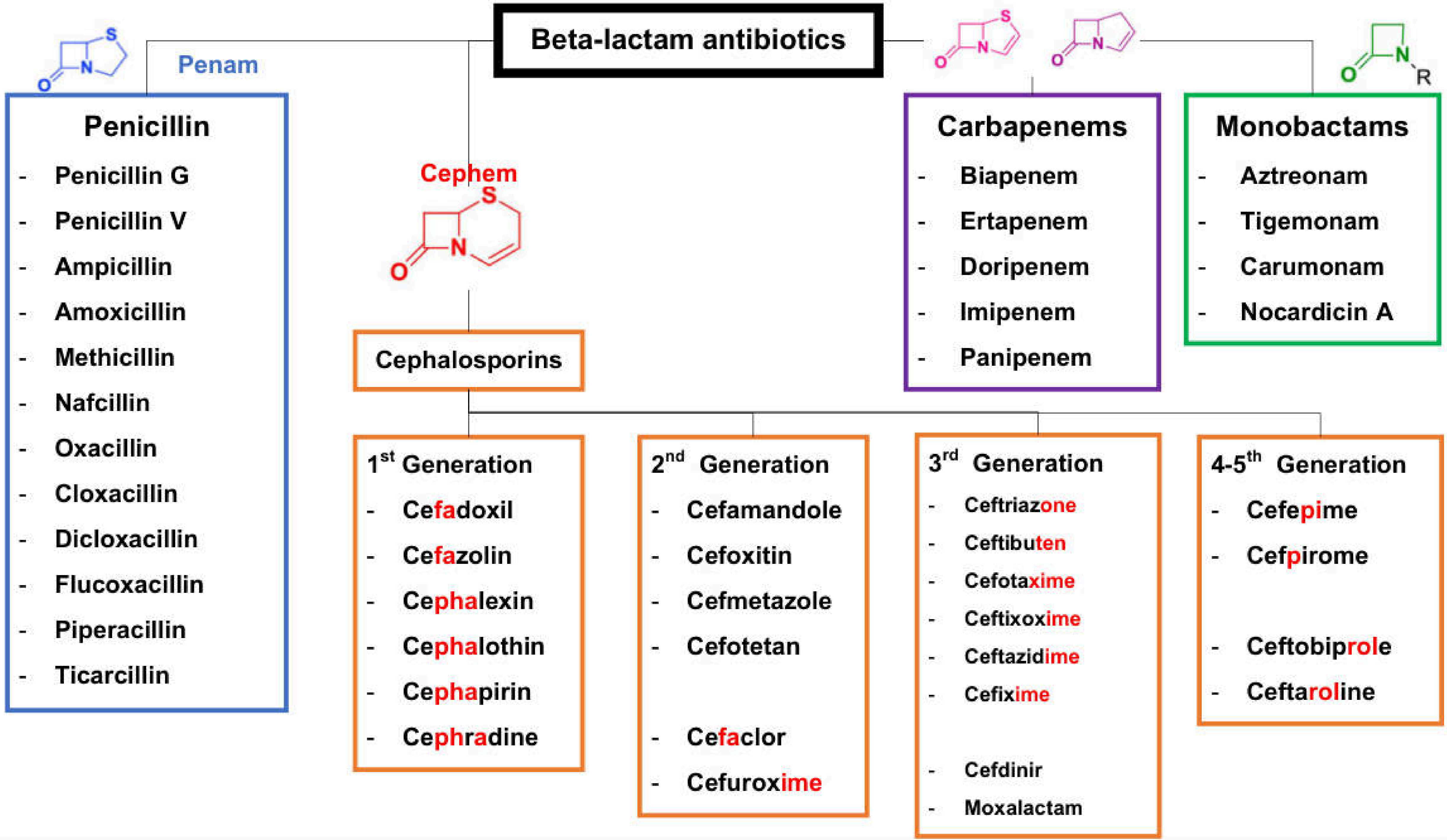
: **Penicillin G (IV): 60-90%**

: **Aminopenicillins (Ampicillin (IV): 90%)**

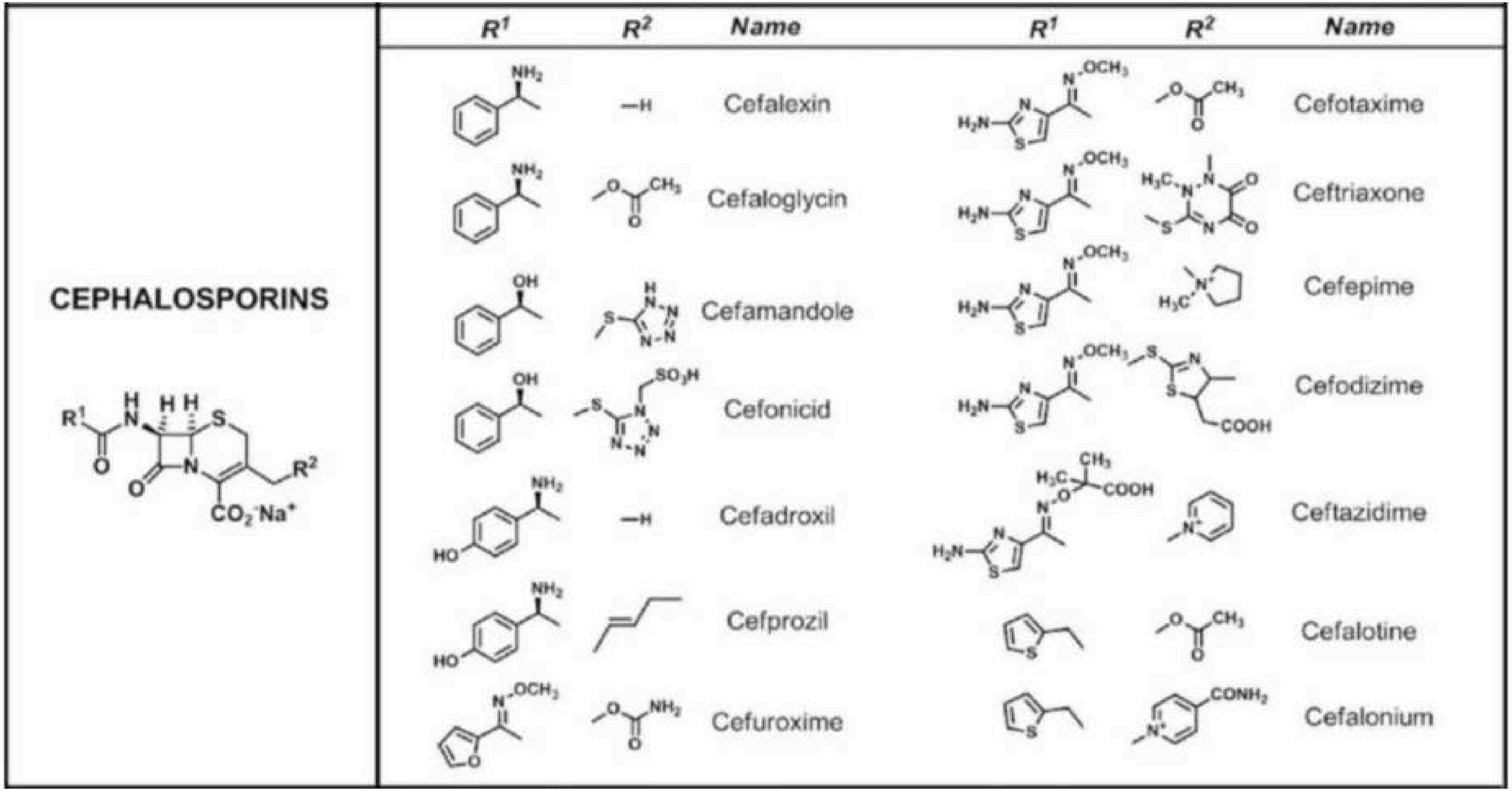
: **Penicillinase-resistance penicillin (Methicillin: 80%, Dicloxacillin: 60%)**

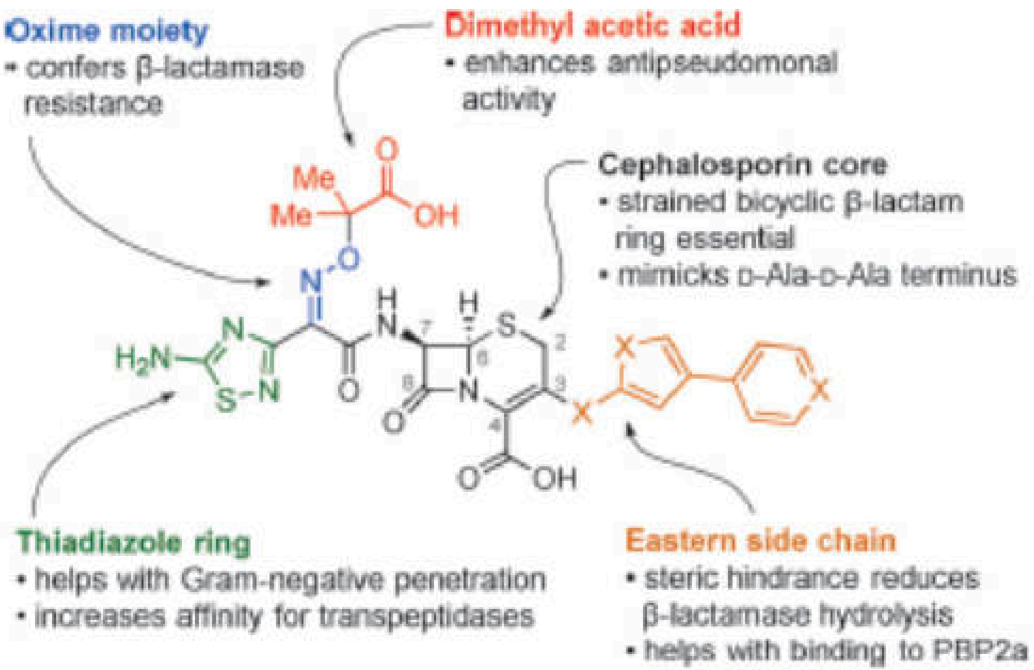
: **Anti-pseudomanal penicillins (Carbenicillin: 85-95%, Ticarcillin: 75-85%, Piperacillin: 60-80%)**

- **Probenecid** blocks the renal tubular secretion of Penicillins → prolong duration of action of Penicillins

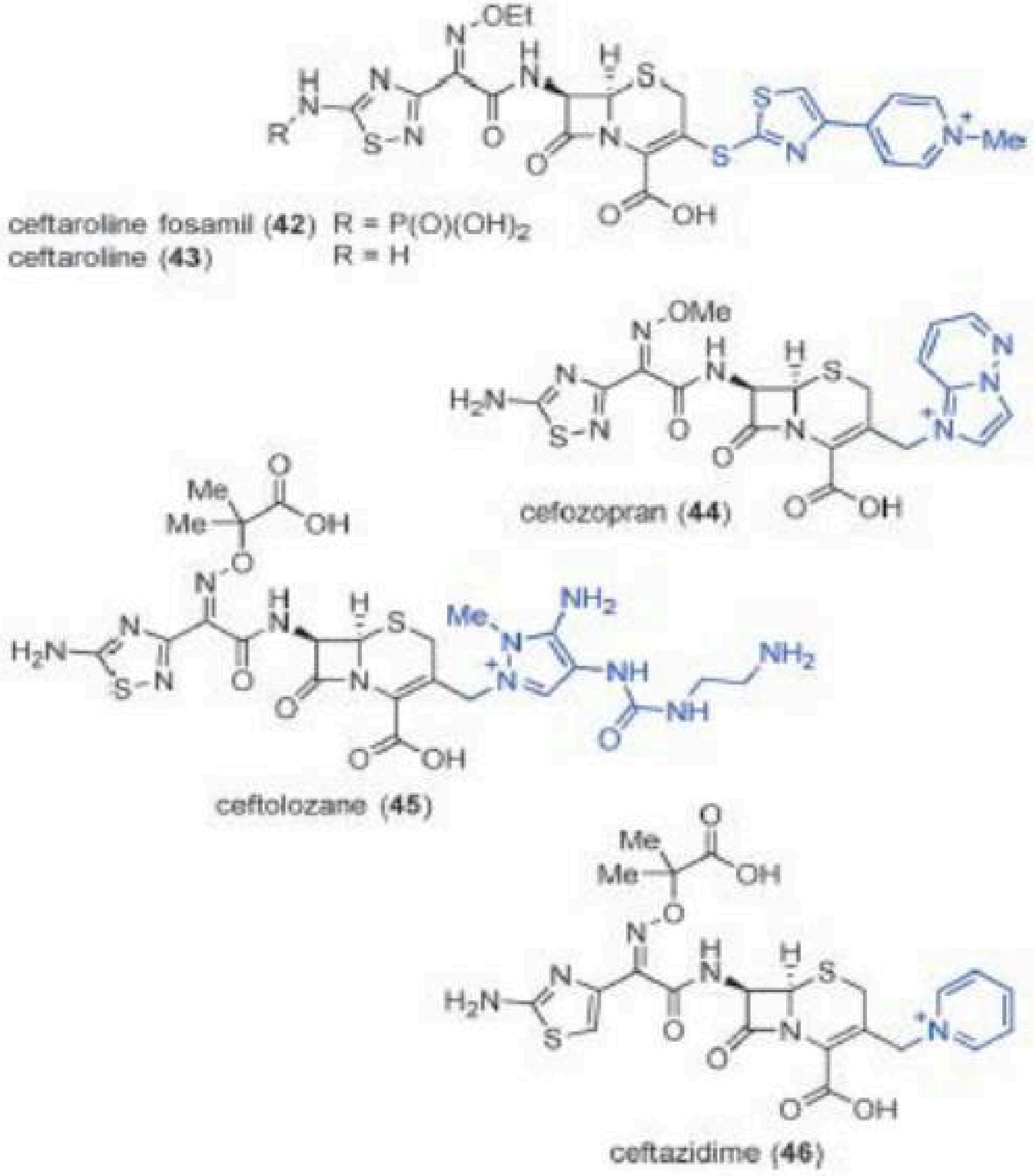


Beta-lactam antibiotics
Penam
Penicillin
- Penicillin G
- Penicillin V
- Ampicillin
- Amoxicillin
- Methicillin
- Nafcillin
- Oxacillin
- Cloxacillin
- Dicloxacillin
- Flucoxacillin
- Piperacillin
- Ticarcillin
Cephem
Cephalosporins
1st Generation
- Cefadoxil
- Cefazolin
- Cephalexin
- Cephalothin
- Cephapirin
- Cephradine
2nd Generation
- Cefamandole
- Cefoxitin
- Cefmetazole
- Cefotetan
- Cefaclor
- Cefuroxime
3rd Generation
- Ceftriazone
- Ceftibuten
- Cefotaxime
- Ceftixoxime
- Ceftazidime
- Cefixime
- Cefdinir
- Moxalactam
4-5th Generation
- Cefepime
- Cefpirome
- Ceftobiprole
- Ceftaroline
Carbapenems
- Biapenem
- Ertapenem
- Doripenem
- Imipenem
- Panipenem
Monobactams
- Aztreonam
- Tigemonam
- Carumonam
- Nocardicin A


CEPHALOSPORINS
R1
R2
Name
Cefalexin
Cefaloglycin
Cefamandole
Cefonicid
Cefadroxil
Cefprozil
Cefuroxime
Cefotaxime
Ceftriaxone
Cefepime
Cefodizime
Ceftazidime
Cefalotine
Cefalonium



Summary of cephalosporin structure–activity and structure property relationships.



Mathieu F. Chellat., et al. Angew. Chem. Int. Ed. 2016, 55, 6600 – 6626

Cephalosporins: Spectrum of Action

- **1st generation cephalosporins (C1G):** Narrow spectrum; good Gram-positive activity and relatively modest Gram-negative activity. Inactivated by Gram-neg beta-lactamases (Derived from *Cephalosporium acremonium*).
- **2nd generation cephalosporins (C2G):** Better Gram-negative coverage (more beta-lactamase stability), but less *Staphylococcal* activity.
- **Cephamycins:** Remain susceptible in presence of Extended Spectrum ß-lactamase (ESBL) because they are not a substrate for the enzyme. Can be used as an indicator for ESBL (Derived from *Streptomyces lactamdurans*).
    - AstraZeneca discontinued its Cefotetan products in June 2006. Currently available from other manufacturers.

Cephalosporins: Spectrum of Action

- **3rd generation cephalosporins (C3G):** Wider spectrum of action when compared to C1G and C2G. Less active than narrow spectrum agents against Gram-positive cocci, but much more active against the *Enterobacteriaceae* and *Pseudomonas aeruginosa* (better beta-lactamase stability).
- **4th generation cephalosporins (C4G):** Broadest spectrum of action. Active against high level cephalosporinases of *Enterobacteriaceae* and *Pseudomonas aeruginosa* (not usually used with ESBL producing organisms).

None have activity to MRSA or *Enterococcus spp*.

- **Next generation - Ceftobiprole :** Broad spectrum; active against the common Gram-negative bacteria. Some Gram-positive activity (Drug Resistant *S. pneumoniae*). Notable for activity against MRSA, unlike any other beta-lactam antibiotic. Bactericidal.

Cephalosporins: Spectrum of Action

	1st Generation *Cefazolin, cephalexin*	2nd Generation *Cefotetan*	3rd Generation *Ceftriaxone, Cefpodoxime Ceftazidime*	4th Generation *Cefepime*	5th Generation *Ceftaroline*
Gram Positives (excl. Enterococci)	++	++	++	++	++
Gram negatives	+	++	++	++	++
Anaerobes	+/-	++	+	+	?
Pseudomonas	-	-	Ceftazidime only	++	-
MRSA	-	-	-	-	++
Primary Coverage	Gram positives *(except MRSA)*	Anaerobes	Gram negatives *(only ceftazidime treats Pseudomonas)*	Pseudomonas	MRSA
Secondary Coverage	None	Gram positives *(except MRSA)* Gram negatives *(except Pseudomonas)*	Gram positives *(except MRSA)*	Gram positives *(except MRSA)* Gram negatives	Other gram positives Gram negatives

Cephalosporins: Pharmacokinetics

- Well distributed in body fluids
- Achieved therapeutic level in CSF: **3rd gen** – **Ceftriaxone, Cefotaxime**
- All cephalosporins cross placenta and secreted in breast milk
- 80 – 90% excreted unchanged in urine
  - <u>except</u> **Cefoperazone** and **Ceftriaxone** are mainly excreted through bile → <u>**No**</u> dose adjustment in renal insufficiency

Cephalosporins: Clinical use

Drug	Clinical use or target pathogens	Recommended dose	Important ADRs
1st generation Cefazolin (IV) Cephalexin (PO)	- Streptococci, MSSA	IV: 1-2 g q 8h PO: 0.25-1 g tid-qid	- Pseudocholethiasis - Kernicterus in newborn
2nd generation Cefuroxime (IV, PO)	- Streptococci, MSSA - H. influenzae - CA-Enterobacteriacea	IV: 0.75-1.5 g q 8h PO: 0.25-0.5 g q 12 h	
3rd generation Ceftriaxone (IV)	- Streptococci, MSSA - PRSP - CA-Enterobacteriaceae	2 g q 24h (CNS infection q12h)	
Cefotaxime (IV)		1-2 g q 8h	
Cefixime Cefdinir cefditoren		Cefixime: 0.2-0.4g q 12h Cefdinir: 0.3g q 12h Cefditoren: 0.2-0.4g q 12h	

Cephalosporins : Clinical use

Drug	Clinical use or target pathogens	Recommended dose	Important ADRs
3rd generation- anti-pseudomonal			
Ceftazidime (IV)	- CA-Enterobacteriaceae - P. aeruginosa - B. pseudomallei - Poor activity against GP	1-2 g q 8h	- Seizure
Cefoperazone/sulbactam (IV)	- Enterobacteriaceae - P. aeruginosa - A. baumannii	1 g cef/0.5-1 g sul Q 8-12h	-Coagulopathy

Cephalosporins : Clinical use

Drug	Clinical use or target pathogens	Recommended dose	Important ADRs
Ceftaroline (IV)	- GPC including MRSA - CA-Enterobacteriaceae - No activity against enterobacterococci	600 mg q 8-12h	- Positive comb's test - Eosinophilic pneumonia
Ceftolozane-tazobactam (IV)	- Enterobacteriaceae including some ESBL - P. aeruginosa - A. baumannii including some MDR strains	1.5 g q 8h	- Renal impairment
Ceftazidime/avibactam (IV)	- Enterobacteriaceae including some ESBL and some CRE - P. aeruginosa - A. baumannii including some MDRstrains	2.5g q 8h	-Seizure (due to ceftazdime) - Hypersensitivity reaction

IC50s of -lactams for *S. pneumoniae* IU1945

β-Lactam class and drug	MIC (μg/ml)	IC_{50} (μg/ml)[a] PBP1a	PBP1b	PBP2x	PBP2a	PBP2b	PBP3	PBP selectivity
Cephalosporins								
Ceftriaxone	0.008	0.15 ± 0.05	0.064 ± 0.006	0.012* ± 0.002	0.21 ± 0.03	>1,000	0.011 ± 0.009[b]	2x, 3[d]
Cefuroxime	0.016	0.22 ± 0.01	0.10 ± 0.02	0.0031 ± 0.0007	0.053 ± 0.005	64 ± 5	0.020 ± 0.000	2x
Cefotaxime	0.016	0.33 ± 0.01	0.22 ± 0.01	0.0078* ± 0.0002	0.17 ± 0.00	>1,000	0.0031 ± 0.0001	2x, 3
Cephalothin	0.25	0.018* ± 0.010	0.013* ± 0.005	0.043 ± 0.005	0.13 ± 0.02	1.9 ± 0.2	0.0060 ± 0.0002	1a, 1b, 3
Cefoxitin	0.5	0.77 ± 0.36	0.14 ± 0.04	0.81 ± 0.24	0.42 ± 0.07	1.6 ± 0.5	0.0020 ± 0.0005	3
Cephalexin	1	>1,000	>1,000	0.67 ± 0.06	>1,000	45 ± 2	0.035 ± 0.001	3
Cefsulodin	4	16 ± 4	2.4 ± 0.2	1.7 ± 0.6	4.3 ± 0.7	>1,000	0.33 ± 0.17	3

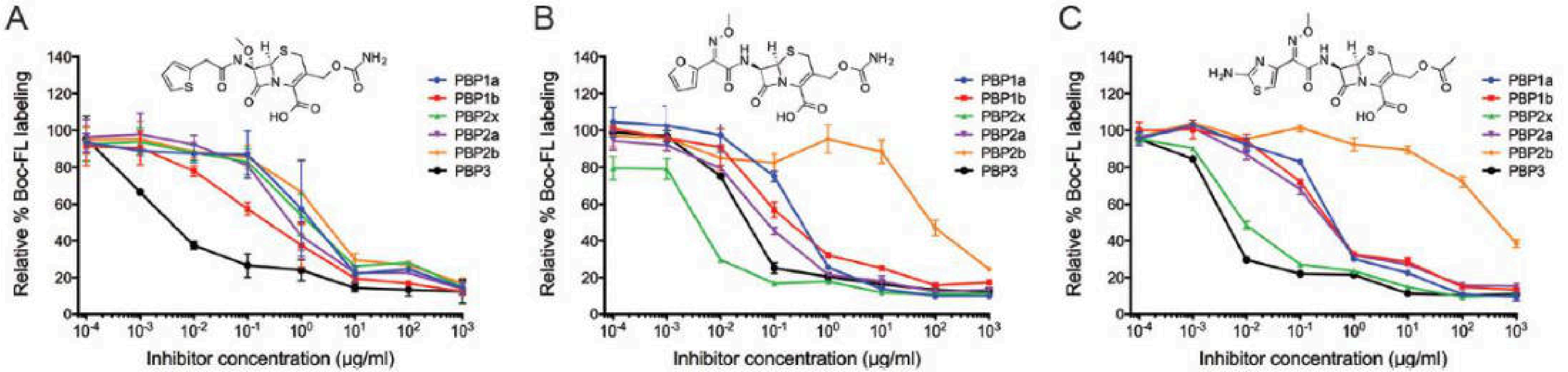


FIG 3 Representative graphs showing the inhibition of PBPs with cefoxitin (A), cefuroxime (B), and cefotaxime (C). Standard deviations for each inhibitor concentration obtained from two independent experiments are plotted. Despite these three molecules' belonging to the same β-lactam class, their selectivity profiles differed substantially, with cefoxitin being selective for PBP3, cefuroxime demonstrating selective inhibition of PBP2x, and cefotaxime inhibiting both proteins (0.01 μg/ml).

กลไกการแพ้ยากลุ่ม penicillin

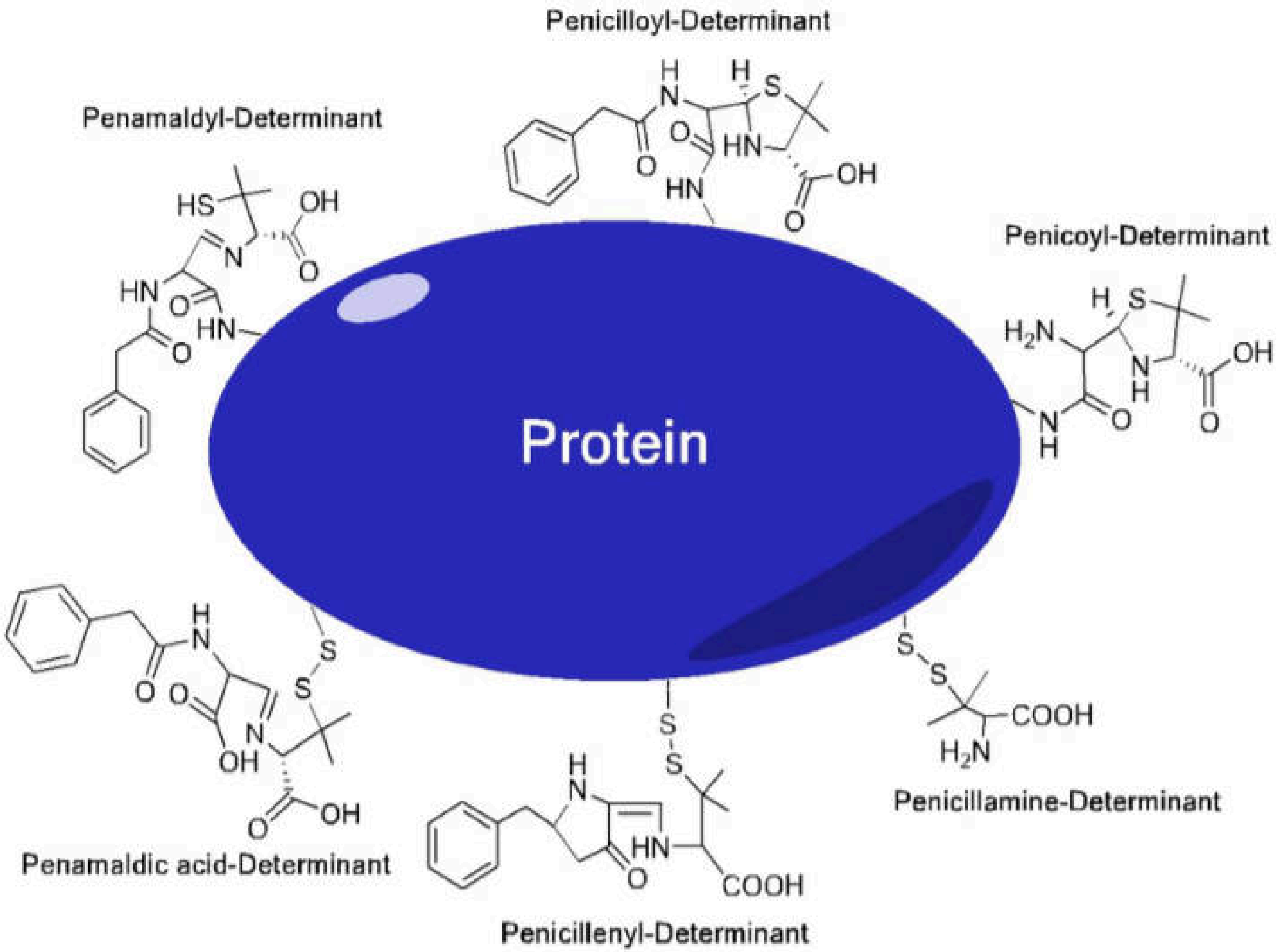
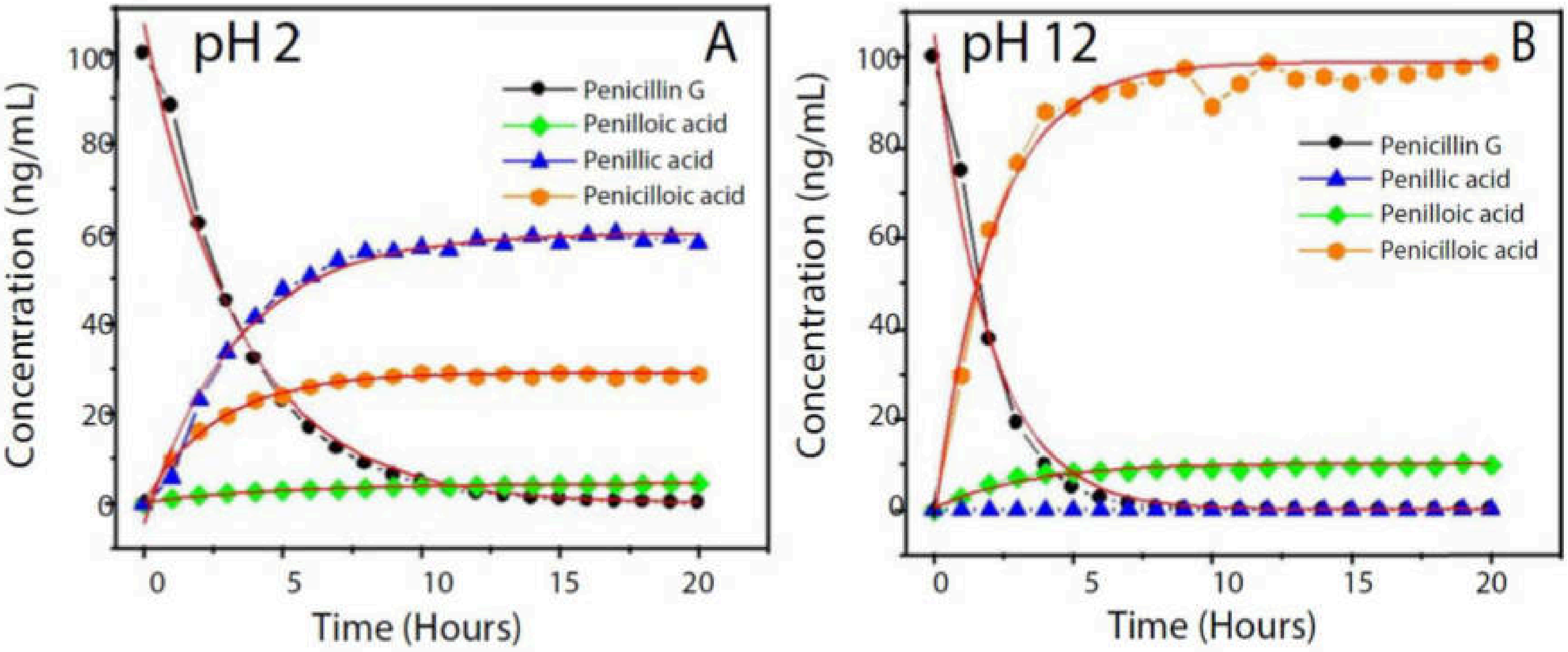
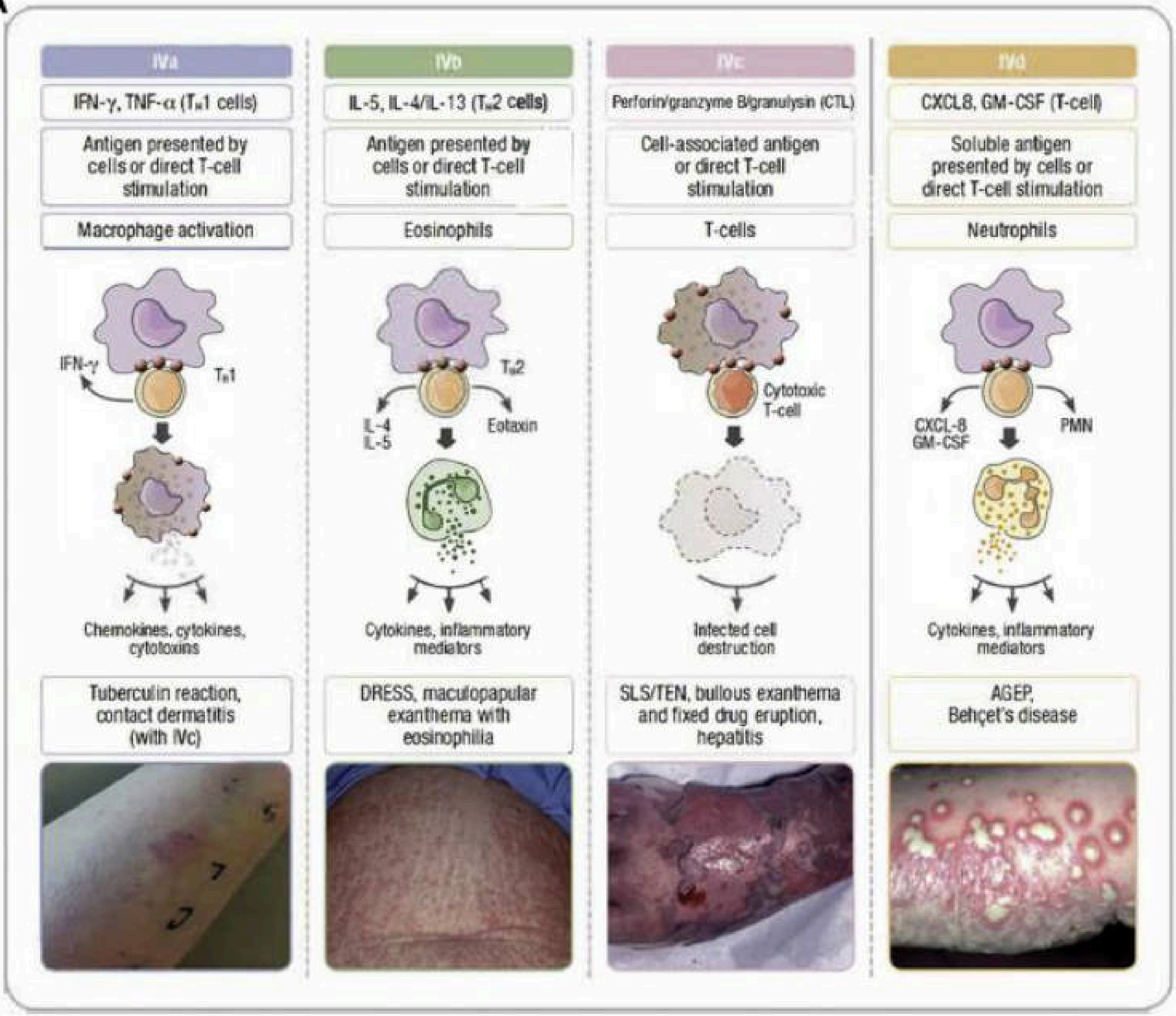


Figure 4. Schematic representation of the chemical structures of metabolite/bio-molecule conjugates causing allergies.

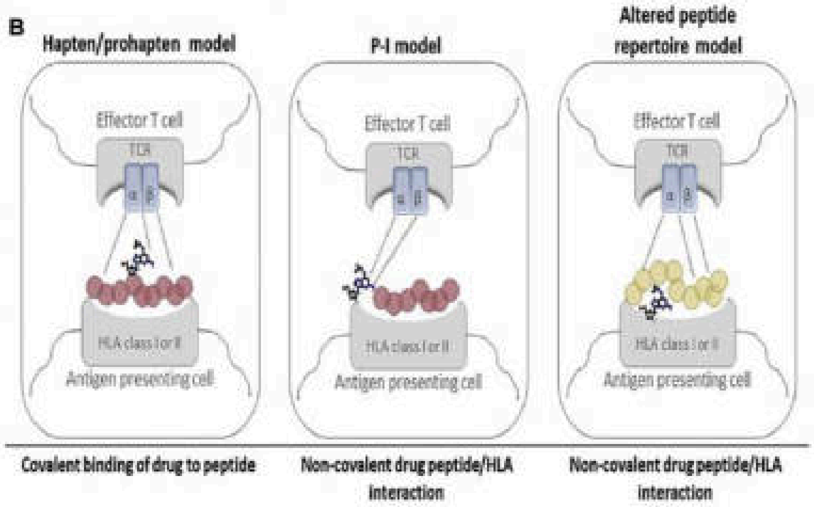


pH 2
A
Penicillin G
Penilloic acid
Penillic acid
Penicilloic acid
Concentration (ng/mL)
Time (Hours)
pH 12
B
Penicillin G
Penillic acid
Penilloic acid
Penicilloic acid
Concentration (ng/mL)
Time (Hours)

A

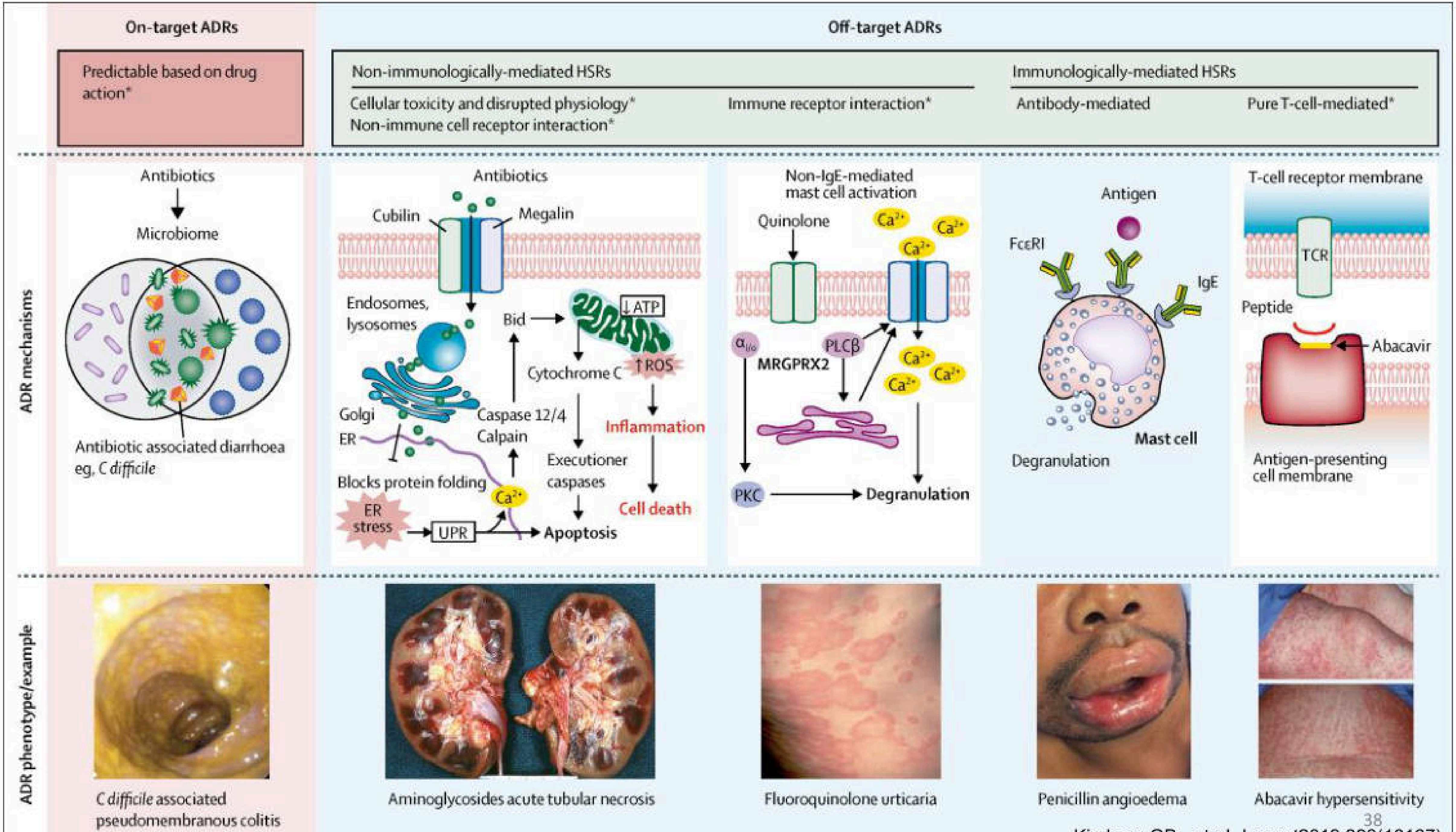


B



Neo-epitope formed by drug binding to peptide	Drug bind to peptide/ HLA at cell surface	Drug binding results In a change in HLA binding motif and a selection of altered peptides

Elizabeth J. Phillips.et al, J ALLERGY CLIN IMMUNOL 2019;143910:66-73



C difficile associated pseudomembranous colitis

Aminoglycosides acute tubular necrosis

Fluoroquinolone urticaria

Penicillin angioedema

Abacavir hypersensitivity

Kimbery GB., et al, Lancet2019,393(10167)

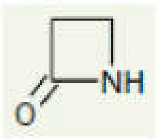
Table 1. β-lactam antibiotics with exact or similar R1 side chains [17,46].

β-Lactam	Agents with Exact or Similar R1 Side Chains	
	Exact R1 Side Chains	Similar R1 Side Chains
Cefaclor	Ampicillin, Cephalexin	Amoxicillin, Cefadroxil, Cefatrizine, Cefmandole, Ceefonicid, Cefprozil, Penicillin G, Penicillin V
Cefadroxil	Amoxicillin, Cefatrizine, Cefprozil	Ampicillin, Cefaclor, Cefmandole, Cefonicid, Cephalexin, Penicillin G, PenicillinV, Piperacillin
Cefatrizine	Amoxicillin, Cefadroxil, Cefprozil	Ampicillin, Cefaclor, Cefmandole, Cefonicid, Cephalexin
Cefazolin	Ceftezole	
Cefditoren	Cefepime, Cefodizime, Cefotaxime, Cefpirome, Cefpodoxime, Ceftriaxone	Ceftaroline, Ceftolozane
Cefepime	Cefditoren, Cefotaxime, Cefozidime, Cefpirome, Cefpodoxime, Ceftriaxone	Ceftaroline, Ceftolozane
Cefiderocol	Aztreonam, Ceftazidime	Ceftaroline
Cefixime		Ceftaroline, Ceftolozane
Cefmandole	Cefonicid, Cefprozil	Amoxicillin, Ampicillin, Cefaclor, Cefadroxil, Cefatrizine, Cephalexin, Penicillin G, Penicillin V, Piperacillin
Cefodizime	Cefditoren, Cefepime, Cefotaxime, Cefpirome, Cefpodoxime, Ceftriaxone	Ceftaroline, Ceftolozane
Cefonicid	Cefmandole	Amoxicillin, Ampicillin, Cefaclor, Cefadroxil, Cefatrizine, Cefprozil, Cephalexin, Penicillin G, Penicillin V, Piperacillin

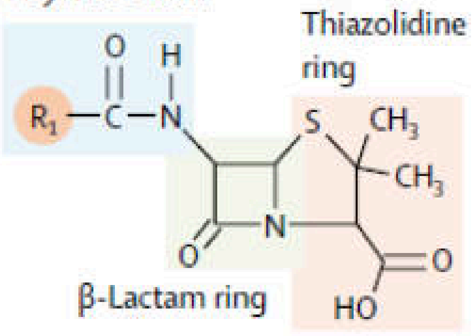
β-Lactam	Agents with Exact or Similar R1 Side Chains	
	Exact R1 Side Chains	Similar R1 Side Chains
Cefotaxime	Cefditoren, Cefepime, Cefodizime, Cefpirome, Cefpodoxime, Ceftriaxone	Ceftaroline, Ceftolozane
Cefoxitin	Cephalothin	Cephalothin
Cefpirome	Cefditoren, Cefepime, Cefodizime, Cefotaxime, Cefpodoxime, Ceftriaxone	Ceftaroline
Cefpodoxime	Cefditoren, Cefepime, Cefodizime, Cefotaxime, Cefpirome, Ceftriazone	Ceftaroline, Ceftolozane
Cefprozil	Amoxicillin, Cefadroxil, Cefatrizine	Ampicillin, Cefamandole, Cefonacid, Cephalexin
Ceftaroline		Cefditoren, Cefepime, Cefodizime, Cefotaxime, Cefpirome, Cefpodoxime, Ceftazidime, Ceftriaxone
Ceftazidime	Aztreonam, Cefiderocol	Ceftaroline
Ceftolozane		Cefpodoxime, Ceftriaxone, Cefixime, Cefotaxime, Cefodizime, Cefepime, Cefditoren
Ceftezole	Cefazolin	
Ceftriaxone	Cefditoren, Cefepime, Cefodizime, Cefotaxime, Cefpirome, Cefpodoxime	Ceftaroline, Ceftolozane
Cephalexin	Ampicillin, Cefaclor	Amoxicillin, Cefadroxil, Cefamandole, Cefatrizine, Cefonicid, Cefprozil, Penicillin G, Penicillin V
Cephalothin	Cefoxitin	

Basic structures

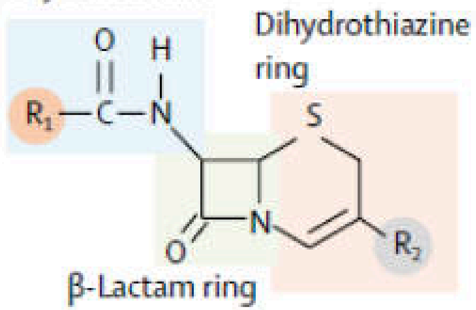
β-Lactam ring



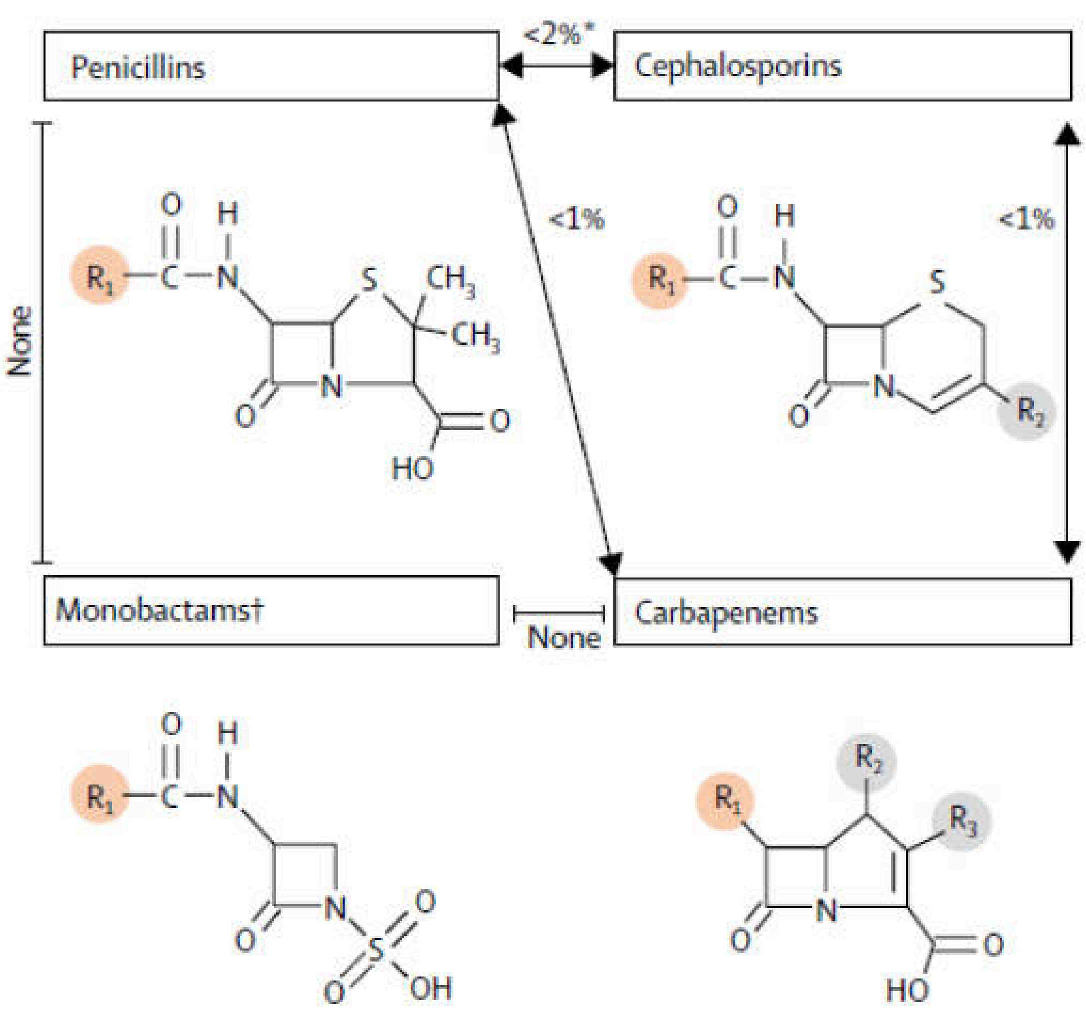
Penicillin structure



Cephalosporin structure



β-Lactam structures and rates of cross-reactivity



Clinically relevant cross-reactivity

Similar side-chains penicillins (R1):

- Penicillin VK and penicillin G

Shared side-chains, penicillins, and cephalosporins (R1):

- Amoxicillin‡ and cefadoxil, cefprozil, cefatrizine
- Ampicillin† and cefaclor, cephalexin, cephradine, cephaloglycin

Shared side-chains cephalosporins (R1):

- Cefadroxil, cefprozil, cefatrizine
- Cefaclor, cephalexin, cephradine, cephaloglycin
- Cefepime, ceftriaxone, cefotaxime, cefpodoxime, ceftizoxime
- Ceftazidime and aztreonam

No shared side-chains, penicillins, and cephalosporins (R1):

- Cefazolin

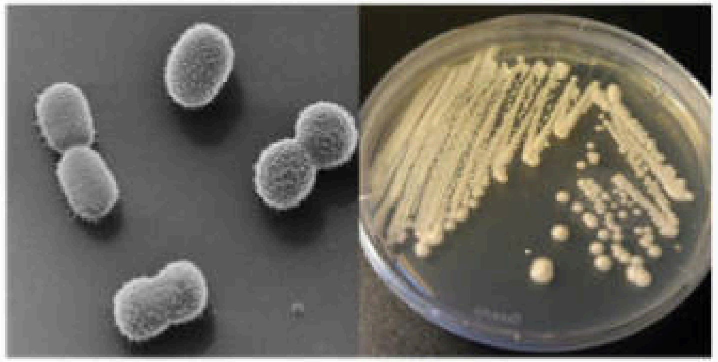
Kimberly G Blumenthal. Et al., the lancet 2019; 393

Table 2. β-lactam antibiotics with exact or similar R2 side chains † [17,46].

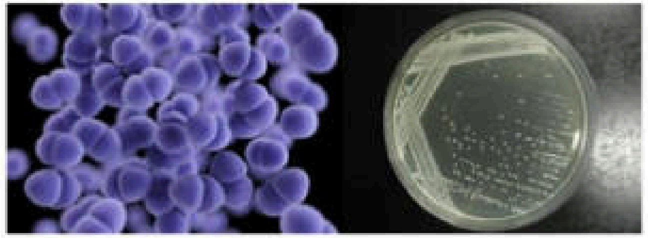
β-Lactam	Agents with Exact or Similar R2 Side Chains	
	Exact R2 Side Chain	Similar R2 Side Chain
Cefazolin		Ceftezole
Cefdinir	Cefixime	
Cefepime		Cefiderocol
Cefiderocol		Cefepime
Cefixime	Cefdnir	
Cefmandole	Cefoperazone, Cefotetan	Cefonici
Cefonicid		Cefmandole, Cefoperazone, Cefotetan
Cefotaxime	Cephalothin, Cephapirin	Cefuroxime
Cefoxitin	Cefuroxime	Cefotaxime, Cefoxitin, Cephapirin
Cefoperazone	Cefamandole, Cefoperazone	Cefonicid
Cefpirome		Ceftazidime
Cefotetan	Cefmandole, Cefoperazone	Cefonicid
Ceftezole		Cefazolin
Cefuroxime	Cefoxitin	Cefotaxime, Cephalothin, Cephapirin
Cephalothin	Cephapirin	Cefoxitin, Cefuroxime
Cephapirin	Cephalothin	Cefoxitin, Cefuroxime

† The R2 side chain cross-reactivity is thought to play a less significant role than R1 side chain cross-reactivity in constituting an immune response.

Multi-drug resistant bacterial pathogens



- Acinetobacter baumannii
  - Gram negative bacteria (aerobic)
  - Isolated from environment, also in hospital
  - Opportunistic infection
    - Immunocompromising patient
    - ICU patient
  - Group of ESKAPE (group of pathogens with high antibiotic resistance rate)
    - Majority of nosocomial infections

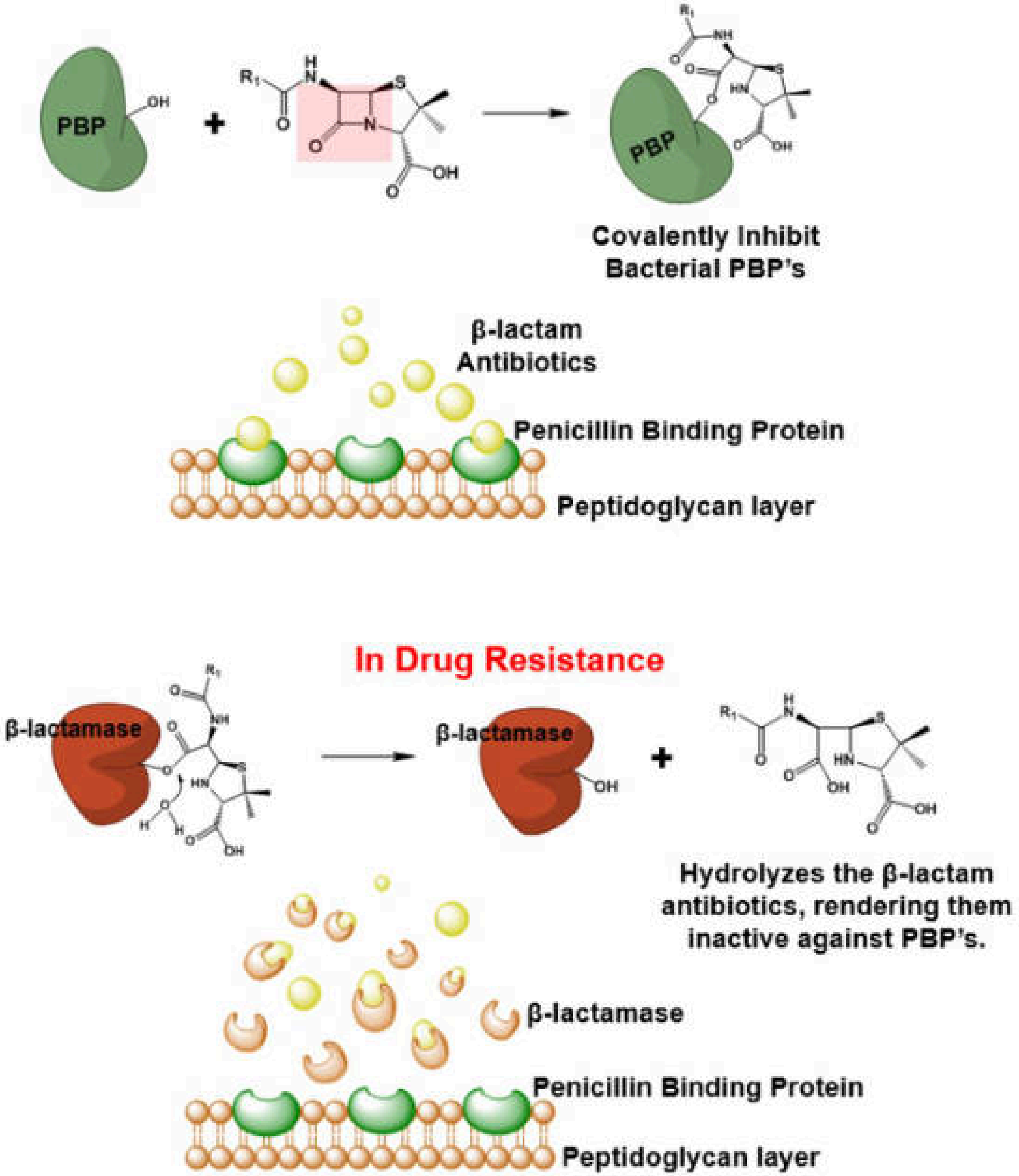


- Methicillin-resistant Staphylococcus Aureus (MRSA)
  - Gram positive bacteria (aerobic)
    - Cause life-threatening bloodstream infections, pneumonia, and surgical site infections
  - Not more virulent than MSSA (Methicillin-sensitive Staphylococcus Aureus), but difficulty treat
    - Cause abscesses, respiratory infections such as sinusitis, and food poisoning
  - Changed PBP, decreased affinity to beta-lamtams
  - Group of ESKAPE (group of pathogens with high antibiotic resistance rate
    - Majority of nosocomial infections

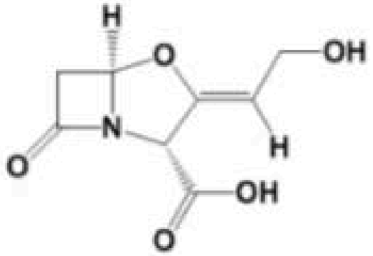
Classification of beta-lactamase

Ambler Class	Representative enzyme types	Examples of enzymes
Class A	Penicillinases Carbapenemases Cephalosporinases Extended-Spectrum β-Lactamases Broad-spectrum-β-Lactamases	CTX-M, TEM, SHV, KPC
Class B	Metallo-β-lactamases	IMP, VIM, NDM-1
Class C	Penicillinases Cephalosporinases	AmpC
Class D	Oxacillin hydrolyzing enzymes	OXA

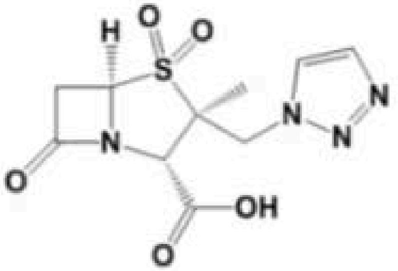
CTX-M: cefotaximase-M, **SHV**: sulfhydryl variable enzymes, **KPC**: *Klebsiella pneumoniae* carbapenemases, **IMP**: imipenemase metallo-β-lactamases, **VIM**: Verona integron encoded metallo-β-lactamases, **NDM-1:** New Delhi metallo-β-lactamase-1, **OXA**: oxacillin hydrolyzing enzymes.



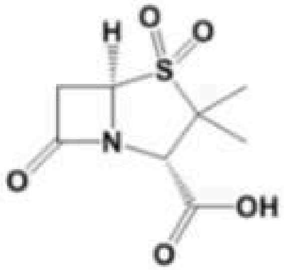
FDA-approved combination therapy against beta-lactam drug resistance



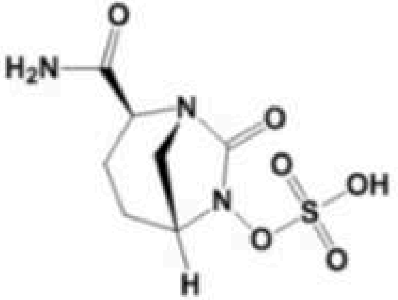
Clavulanic Acid
(+) Amoxicillin



Tazobactam
(+) Piperacillin

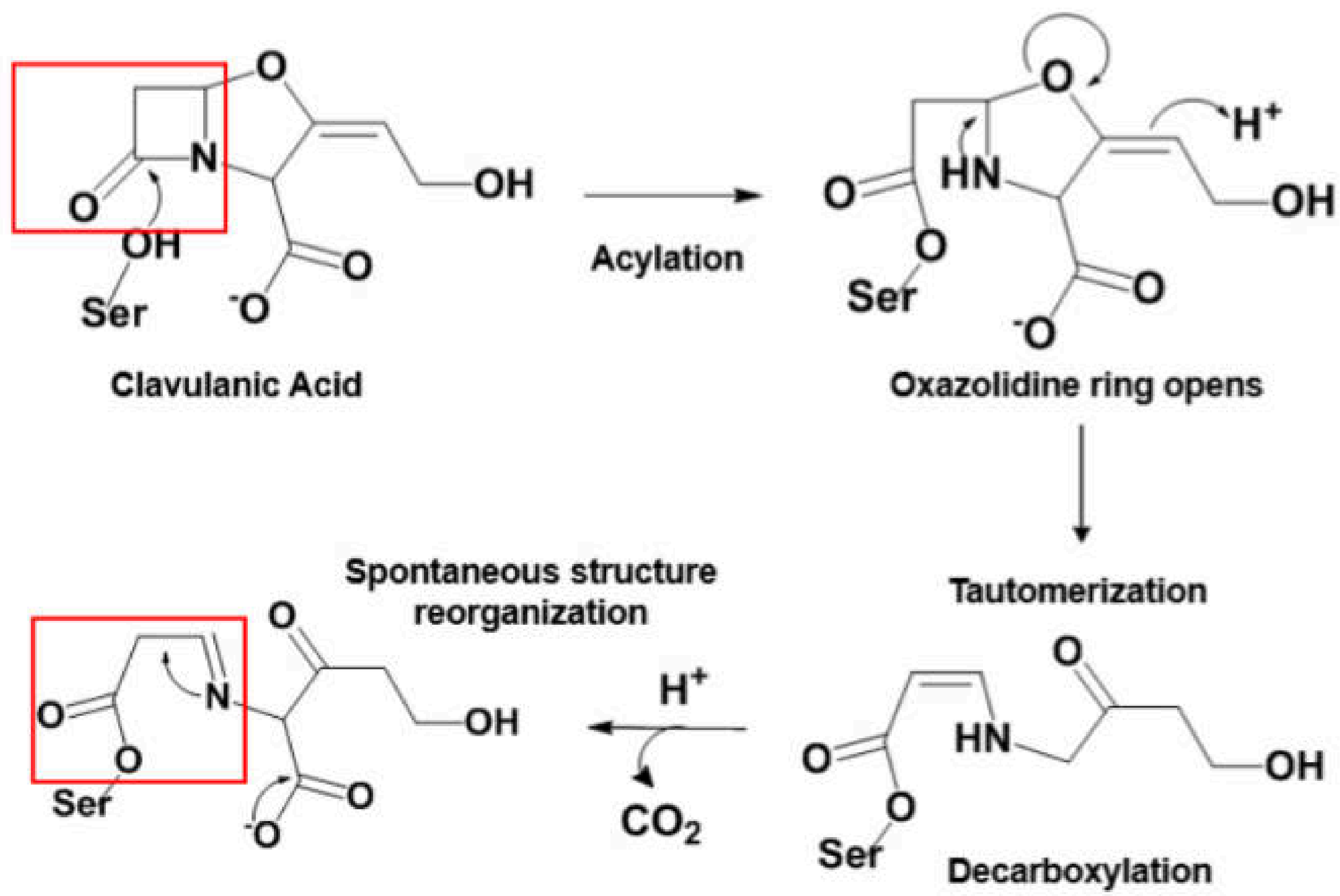


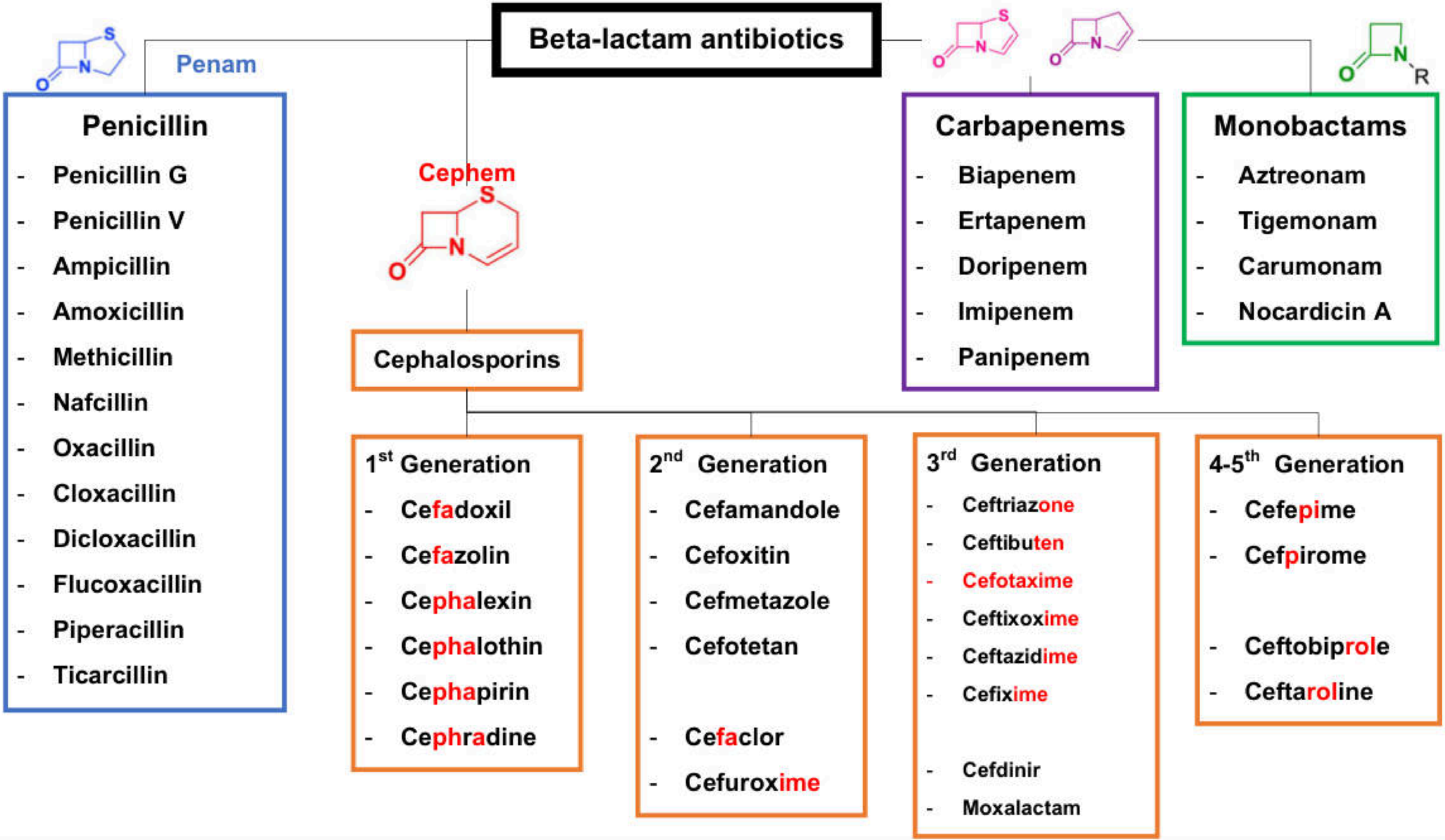
Sulbactam
(+) Ampicillin



Avibactam
(+) Ceftazidime

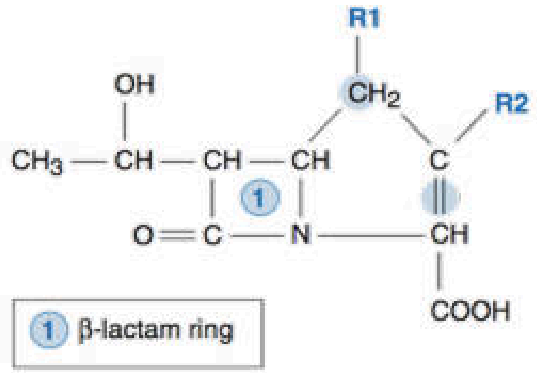
Mechanism of beta-lactamase inhibitor as an irreversible suicide inhibitor



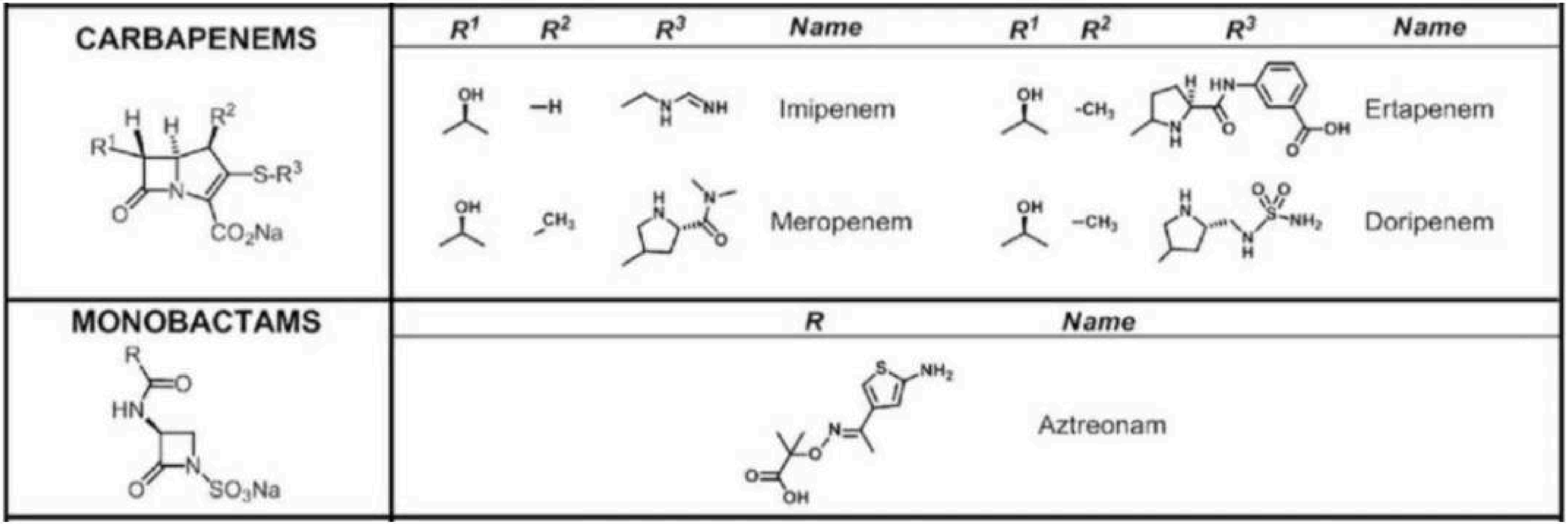

Beta-lactam antibiotics
Penam
Penicillin
- Penicillin G
- Penicillin V
- Ampicillin
- Amoxicillin
- Methicillin
- Nafcillin
- Oxacillin
- Cloxacillin
- Dicloxacillin
- Flucoxacillin
- Piperacillin
- Ticarcillin
Cephem
Cephalosporins
1st Generation
- Cefadoxil
- Cefazolin
- Cephalexin
- Cephalothin
- Cephapirin
- Cephradine
2nd Generation
- Cefamandole
- Cefoxitin
- Cefmetazole
- Cefotetan
- Cefaclor
- Cefuroxime
3rd Generation
- Ceftriazone
- Ceftibuten
- Cefotaxime
- Ceftixoxime
- Ceftazidime
- Cefixime
- Cefdinir
- Moxalactam
4-5th Generation
- Cefepime
- Cefpirome
- Ceftobiprole
- Ceftaroline
Carbapenems
- Biapenem
- Ertapenem
- Doripenem
- Imipenem
- Panipenem
Monobactams
- Aztreonam
- Tigemonam
- Carumonam
- Nocardicin A

Carbapenems

Parenteral Agents	Oral Agents
Imipenem/cilastatin	None
Meropenem	
Doripenem	
Ertapenem	



- These molecules are quite small and have charge characteristics that allow them to utilize special porins in the outer membrane of gram-negative bacteria to gain access to the PBPs.
- The structures of the carbapenems make them resistant to cleavage by most B-lactamases.
- The carbapenems have an affinity for a broad range of PBPs from many different kinds of bacteria.



CARBAPENEMS
R1 R2 R3 Name
Imipenem
Ertapenem
Meropenem
Doripenem
CO2Na
S-R3
MONOBACTAMS
R Name
Aztreonam
SO3Na

Carbapenems: Spectrum of Action

Drug	Strep spp. & MSSA	Entero-bacteriaeae	Non-fermentors		Anaerobes
			Pseudomonas	Acinetobacter	
Imipenem	+	+	+	++	+
Meropenem	+	+	++	+	+
Ertapenem	+	+	Limited activity		+
Doripenem	+	+	++	++	+

Not active against *Stenophomonas, Flavobacterium, MRSA*
Not active against *Chlamydia, Mycoplasma,* or *Legionella*

Imipenem

- Broad spectrum
  - Gram +
  - Gram – aerobic
    - Enterobacteriaceae
    - *P. aeruginosa*
  - Anerobic
- Binding with PBP (PBP2 and PBP1a)
- ทนต่อ beta-lactamase แต่ไม่ทนต่อ Dehydropeptidase-I (DHP-I) ที่พบในเนื้อเยื่อของร่างกาย
  - จึงใช้ร่วมกับ Cilastatin (IV)
  - ปรับยาในผู้ป่วยโรคไต
- เป็นยาที่สงวนไว้ใช้ในกรณีติดเชื้อที่รุนแรงเท่านั้น
- มีการกระจายตัวของยาได้ดีมาก จึงใช้ได้ในการติดเชื้อที่อวัยวะต่าง ๆ ยกเว้นการติดเชื้อที่ถุงน้ำดี
- พิษต่อระบบประสาท ก่อให้เกิดอาการชัก โดยเฉพาะอย่างยิ่งในผู้ป่วยที่มีรอยโรคในสมอง จึงไม่แนะนำให้ใช้ในผู้ป่วยเยื่อหุ้มสมองอักเสบ

Meropenem

- ทนต่อ dehydropeptidase-I (DHP-I)
- ใช้กับ P. aeruginosa ที่ดื้อ imipenem ได้
- มีความไวต่อ Gram+ cocci น้อยกว่า imipenem
- PK ไม่แตกต่างกันระหว่างการให้ยาแบบ IV และ IM
- PK ไม่แตกต่างกันด้วยปัจจัยทางด้านขนาดยาและอายุ
- ใช้ในติดเชื้อในกระแสเลือด เยื่อหุ้มสมองอักเสบ ไส้ติ่งแตก
- เป็นยาที่สงวนไว้ในกรณีที่ติดเชื้อรุนแรง

Carbapenems : Clinical use

Drug	Clinical use or target pathogens	Recommended dose	Important ADRs
Ertapenem (IV)	- Enterobacteriaceae including most ESBL	1 g q 24 h	- Dizziness, confusion
Imipenem/Cilastatin (IV)	- Enterobacteriaceae including ESBL - P. aeruginosa - A. baumannii including MDR strains - Enterococci (only imi/cilas)	500 mg q 8h	- Seizure
Meropenem (IV)		1 g q8h (CNS infection; 2 g q 8h)	-Drug interaction with valproic acid

Monobactams & Carbapenems: Spectrum of Action

Beta-lactams: Monobactams

Spectrum of Action

- **Aztreonam:** Gram-negatives (*Enterobacteriaceae* and *Pseudomonas*). Not hydrolyzed by most commonly occurring plasmid and chromosomally mediated ß-lactamases, and does not induce the production of these enzymes.